# Reject, Punish, or Withdraw? Cold and the Political Response of Highland Households in Peru*

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**Abstract**

When a shock reveals that the state does not protect them, citizens may reject the political system, punish those in office, or withdraw. We ask which response prevails in the Peruvian highlands, using exposure to cold as the shock. We match twelve rounds of the national household survey to hourly temperatures at each household's location and identify the effects of cold from the variation in the timing of interviews within districts. Cold lowers farm revenue, kills livestock, and makes vulnerable household members more likely to fall ill and require medical attention. Households conclude that democracy delivers less: they become less likely to believe that democracy works well and that the rights it promises are respected, while support for democracy as a principle and confidence in specific institutions do not change. They withdraw rather than punish: absenteeism rises in presidential and municipal elections despite mandatory voting, but incumbent mayors are not punished; participation in community associations rises, and individual identity shifts from the state to indigenous groups. The erosion of support is concentrated where Peru's main cash transfer program reaches few of the poor, suggesting that social protection shields the relationship between citizens and democracy when extreme weather strikes.

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# 1 Introduction

Extreme weather tests the state. Cold can destroy harvests, kill livestock, and affect the health of household members. In light of such adverse events, households learn how much protection the state delivers, and their political attitudes and behavior may change as a result. How citizens respond when the state fails them in a crisis is question that has long generated interest, and the possible answers fall into three broad categories. Citizens may *reject* the political system, losing faith in democracy itself when it fails to deliver (Grosjean et al., 2013). They may *punish* those in office, voting against incumbents or losing confidence in particular institutions (Cole et al., 2012; Gasper and Reeves, 2011). Or they may *withdraw*, staying away from the polls (Eichengreen et al., 2024; Guiso et al., 2024) and potentially turning instead to the local, informal institutions that help them cope. Which response prevails matters. Punishment is the mechanism through which elections discipline governments; rejection and withdrawal weaken democracy without disciplining anyone. Yet the three are hard to distinguish empirically.

In this paper, we study these responses among farming households in the Peruvian highlands, using cold events as adverse shocks. We match the 2007–2018 rounds of the Peruvian National Household Survey (ENAHO) to hourly temperatures from the ERA5 reanalysis of the European Centre for Medium-Range Weather Forecasts, using the approximate GPS location of each household and its date of interview. We measure exposure to cold as the cumulative degree hours (CDH) below -9°C over the 12 months before the interview, a measure that captures both how long and by how much temperatures fell below a threshold that damages the main highland crops. Because the ENAHO is fielded continuously, households in the same district are interviewed in different months and face different weather histories; with district, year, and calendar-month fixed effects, we identify the effect of cold from this within-district variation in interview timing.

Three features make the setting well suited to the question. First, cold is a sharp and exogenous shock that reaches a farming household through several channels at once: crops, livestock, and the health of household members. In the Peruvian highlands, frosts and cold waves have become more frequent (Keller and Echeverría, 2013; FAO, 2008; Tambet and Stopnitzky, 2021). Temperatures are expected to reach almost -20°C in some areas during the 2026–2027 El Niño cycle (Peru 21, 2026). Second, the ENAHO asks the respondents about democracy as a principle, how democracy works

in practice, their confidence in specific institutions, and the groups they identify with; together with district-level voting data from presidential and municipal elections as well as information on individuals' self-reported participation in local associations, it lets us observe rejection, punishment, and withdrawal in a single population. Third, Peru is a democracy with weak parties, a long history of corruption and instability, and some of the lowest levels of political trust in Latin America (Bargsted et al., 2017).

We first establish that cold is a multifaceted adverse shock for the household. A one standard deviation increase in CDH below -9°C lowers annual agricultural revenue by 1.6 percent and increases the value of livestock deaths by 2.9 percent. Cold in the weeks before the interview raises the probability that a child under five or an adult over 65 needs medical attention or is hospitalized. Cold exposure in the prior year increases the likelihood that households report suffering a natural disaster, having a household member fall gravely ill, and experiencing a problem that cost them income or assets over the same period.

We then examine how cold exposure affects political attitudes. Cold lowers the belief that democracy works well: a one standard deviation increase in CDH below -9°C reduces the probability that a respondent believes democracy works well or very well by 0.44 percentage points. Exposure of 100 degree hours in the past year (about three nights at -12°C or 10°F), the level experienced by households in our sample during severe winters, lowers the belief that democracy works well by nearly 4 percentage points (7 percent relative to the mean). This relationship is robust across different temperature thresholds, exposure windows, and functional forms of the shock, to alternative definitions of the outcome, to finer fixed effects, to randomization inference, and to using an alternative dataset (i.e., from the Latin American Public Opinion Project, LAPOP). Importantly, we find no evidence that cold induces endogenous migration ort affects survey timing, and we also find no evidence of anticipation by households or non-parallel pretrends using a falsification exercise with future cold exposure as a placebo.

To help interpret this result, we turn to perceptions of democratic efficacy, belief in democratic principles, and confidence in government institutions. Cold exposure lowers respondents' view of whether democracy delivers the rights it promises (such as freedom of expression, equality before the law, political freedom, transparent elections). In contrast, cold does not affect individuals' view of democracy as a principle: the shares of respondents who consider democracy important

and preferable to any other form of government do not move. Nor does the shock single out any government institution for blame: confidence in Congress, political parties, regional governments, and provincial and district municipalities is unaffected, as is approval of how each level of government is managed. Confidence in institutions unrelated to the state — the press, radio and television, the Catholic church — is unaffected too, so the decline is not a general souring of mood or a change in how respondents answer surveys. In this context, households hit by cold do not reject democracy in principle and do not punish a particular government; they conclude that democracy is delivering less.

Voting behavior follows the same pattern. Using district-level data from the 2006, 2011, and 2016 presidential elections, we find that cold in the year before an election raises the share of registered voters who do not vote: a one standard deviation increase in district-level CDH raises absenteeism by about 0.3 percentage points in each round, an increase of 1.7 percent relative to the mean and larger than the margin by which the 2016 run-off was decided (0.2 points). Voting is mandatory in Peru and abstention is fined, so this is a costly form of withdrawal. Punishment of an incumbent is not observable in presidential elections because Peruvian law prohibits incumbent candidates from running; moreover, in all three elections, the governing party's candidate did not even participate. Thus, to test for incumbency effects, we turn to municipal elections, in which incumbent mayors could stand for reelection until a 2015 law barred it. In those elections, cold does not reduce the probability that the incumbent mayor runs again or is re-elected, nor the share of the vote they receive, while absenteeism rises in these elections as well. At the same time, households turn toward the community: participation in local associations rises, entirely through community-based associations, and respondents become less likely to identify with their department, province, district or town and more likely to identify with their indigenous group. An alternative dataset (the LAPOP) shows the same shifts in self-reported voting and in indigenous self-identification. Extreme cold thus moves households away from the formal, national institutions of the state and toward local, informal ones.

Finally, we ask whether the response depends on coverage of the state's safety net. We split provinces by the baseline coverage of Juntos, Peru's main conditional cash transfer program — the number of households it paid in 2012 per *poor* resident (to distinguish heterogeneity by coverage from heterogeneity based on poverty) — and allow the effects of cold to differ between provinces

with high and low coverage. The decline in the belief that democracy works well is concentrated in provinces with low coverage and disappears where coverage is high. Cold raises absenteeism in both groups, but less so where coverage is high. Results on turning towards indigenous identity are qualitatively consistent with our democracy perceptions and voting results, though the difference between high- and low-coverage provinces is not statistically significant, while participation in community associations does not differ by coverage. Generally, these results are consistent with the view that where the state reaches households with income support, a shock is less likely to be perceived as a failure of democracy to deliver.

Our paper contributes to three literatures. The first asks how adverse shocks shape citizens' political attitudes and behavior. Thus far, this literature has developed in two strands that rarely meet. One strand studies the determinants of political trust and support for democracy. Within economics, the evidence has focused on exposure to violence, crime, and conflict (Blanco and Ruiz, 2013; Blanco, 2013; Malásquez and Salgado, 2023) and on economic insecurity and unemployment (Guiso et al., 2024; Stevenson and Wolfers, 2011); a broader literature, reviewed by Giuliano and Spilimbergo (2025), shows that aggregate shocks such as recessions, epidemics, and natural disasters leave lasting marks on preferences and beliefs, including trust in institutions.[1] The other strand studies what voters do after a shock, and finds that they hold incumbents to account for natural disasters and for the relief that follows (Healy and Malhotra, 2009; Gasper and Reeves, 2011; Cole et al., 2012; Ahlerup et al., 2024). Studies in the first strand seldom observe behavior, and studies in the second seldom observe what voters conclude about the political system beyond the incumbent. Closest to our study, Balcazar and Kennard (2025) find that unusually high temperatures in India lower confidence in state governments and the police, raise cooperation within communities, and raise turnout, anti-incumbent voting, and protest. We find the same erosion of support and the same turn toward the community, but withdrawal from elections rather than mobilization, and no punishment of municipal mayoral incumbents. Their model predicts mobilization only where citizens hold a grievance against an identifiable incumbent, a condition that Peru's presidential

[1] Outside of economics, a large literature examines the roots of political support, though much of it concerns rich democracies and is correlational (Citrin and Stoker, 2018; Zmerli and Van der Meer, 2017; Zmerli and Castillo, 2015; Rhein and Jansesberger, 2024).

elections do not meet. Our setting also lets us go further on the attitudinal side. The ENAHO allows us to distinguish support for democratic principles, the regime's performance, and confidence in specific institutions and actors; additionally, we trace effects on both electoral outcomes and engagement in local associations. We can therefore ask which of the responses a shock can trigger actually occurs: not only whether voters punish the incumbent, but whether they lose faith in democracy itself, judge that it delivers less, or lose confidence in particular institutions, and whether they stay away from the polls or turn toward local institutions and identities. Our results on heterogeneity by cash transfer coverage relates to work showing that redistributive programs can foster support for government (Evans et al., 2019; Kosec and Mo, 2023; Zucco Jr, 2013) and that vulnerability to shocks shapes the relationship between citizens and the state (Bobonis et al., 2022).

The second literature links weather to conflict and political instability (for example, Miguel et al., 2004; Burke et al., 2009; Dell et al., 2012; Hsiang et al., 2011; Harari and La Ferrara, 2018; Ranson, 2014; Maystadt and Ecker, 2014; Sarsons, 2015; Hsiang et al., 2013).[2] Weather shocks can also change political regimes themselves (Brückner and Ciccone, 2011; Rahman et al., 2022). We document an earlier stage of that process. Weather shocks erode the evaluation of the political system and participation in formal, national institutions, which prior work has identified as precursors of conflict (Engvall, 2010; Buhaug, 2010), and they strengthen local institutions and identities, in line with evidence that communities self-organize when the state and markets fail to insure them (Bhattamishra and Barrett, 2010; Buggle and Durante, 2021).[3]

Third, we add cold to the shocks whose economic and political consequences have been documented in developing countries. Floods, droughts, and extreme heat have been studied extensively (e.g., Dell et al., 2012; Skoufias et al., 2012; Zhang et al., 2017; Schlenker et al., 2009; Aragón et al., 2021; Deschenes and Moretti, 2009; Deschênes and Greenstone, 2011; Balcazar and Kennard, 2025); extreme cold has received much less attention, even though cold-related deaths account for far more global mortality than heat-related deaths (Zhao et al., 2021), cold waves

[2] A related strand studies the long-run consequences of climatic change for conflict in historical societies (Chaney, 2013; Kung and Ma, 2014; Jia, 2014; Yancheva et al., 2007).

[3] Our results on cash transfers also relate to evidence that policy can attenuate the effects of weather shocks on conflict (Fetzer, 2014; Sarsons, 2015; Garg et al., 2020) and that awareness of weak public services depresses turnout (Cox et al., 2024).

continue to strike highland and rural populations in Asia and Latin America (Marengo et al., 2023; Shrestha et al., 2023; United Nations Development Programme (UNDP), 2025), and climate change may make cold extremes more frequent in both hemispheres (Cohen et al., 2018; Kim et al., 2017; Cai et al., 2015; Geng et al., 2023).[4] In related work, Bollman et al. (2026) use the same measure of exposure to show that extreme cold raises intimate partner violence in Peru; here we document its consequences for households' relationship with the state.

The rest of the paper proceeds as follows. Section 2 describes the political setting and the nature of extreme cold in the highlands. Section 3 presents the data and Section 4 the empirical strategy and the threats to identification. Section 5 establishes the effects of cold on the household, Section 6 on perceptions of democracy, and Section 7 on electoral participation, local associations, and identity. Section 8 examines heterogeneity by the baseline coverage of the Juntos cash transfer program, and Section 9 concludes.

## 2 Background

### 2.1 Politics in Peru

Formally, Peru is a democracy in which the president and members of Congress are elected by popular vote every five years. In practice, its political system has been plagued by unstable governments and corrupt leaders over the last three decades; The Economist Intelligence Unit (2023) classifies Peru as a *hybrid regime*, somewhere between a "flawed democracy" and an "authoritarian regime". Political parties collapsed in the 1990s — a phenomenon that Tanaka (2005) describes as having left Peru a "party-less democracy" — and have remained weak since, so that few stable partisan outlets participate in national politics. Every elected president since 1985 has either faced jail time for corruption or has had credible corruption allegations against them (Bristow, 2022), and members of Congress and of the National Board of Justice (the council that appoints and removes judges) have also faced corruption charges (Wall Street Journal, 2020; IDEHPUCP, 2020;

[4] In the northern hemisphere, accelerated Arctic warming can destabilize the polar vortex and push cold air into the mid-latitudes (Cohen et al., 2018, 2021; Hamouda et al., 2024); in the southern hemisphere, cold events are associated with La Niña episodes, which are projected to become more frequent and prolonged (Cai et al., 2015; Geng et al., 2023).

La República, 2021). Since 2016, tensions between the executive and the legislature have escalated: Congress has impeached three presidents, one president dissolved Congress, and another staged a failed *coup d'état*.

These governance problems have coexisted alongside relatively poor provision of public services, including health care and social assistance on which households hit by an adverse shock depend. In theory, the government provides universal access to health services in public facilities. In practice, Aguirre Martens (2023) estimates that seven out of ten people who needed medical care did not receive it, mainly because of long waiting times (35%), and those who do obtain medical consultations tend to be the wealthiest (Díaz Ruiz et al., 2024). The flagship social assistance program is Juntos, a conditional cash transfer launched in 2005 that provides poor families with a monthly subsidy of 100 soles (about US$29). The program was rolled out by district according to a poverty index. In the highland districts we study, the share of districts where Juntos paid any household rose from 5% in 2005 to 46% in 2007, rose again to 66% in 2012 and 79% by 2017 (Ministerio de Desarrollo e Inclusión Social (MIDIS) and Programa Nacional de Apoyo Directo a los más Pobres JUNTOS, nd). Juntos has improved outcomes such as early nutritional status and school attendance (see Sánchez and Rodríguez 2016 for a synthesis), but its nominal amount has not changed since 2010, and inflation has eroded the real value of a transfer that represented about 10% of beneficiary households' consumption in the early years of the program (Scott et al., 2022).

It is thus not surprising that Peru has one of the lowest levels of political trust in Latin America. In a sample of 17 Latin American countries, Peruvians' trust in democratic institutions (the national congress, political parties, and the government) is the second-lowest, surpassing only Ecuador (Appendix Figure A1), and only half of Peruvians believe that democracy is better than alternative systems of government (Appendix Figure A2).

Presidential elections typically include two rounds: all parties compete in the first round, and if no candidate wins more than 50% of the vote, the top two compete in a run-off. Voting is mandatory for all citizens aged 18–70, and those who do not cast a vote face a fine that ranges from approximately US$6 (in poorer districts) to approximately US$25 (in wealthier districts). Because of these penalties, turnout is generally high but not universal; for example, in 2011, turnout in the second round was 82.5% (European Union Election Observation Mission, 2011).

## 2.2 Extreme cold in the Peruvian highlands and beyond

In this paper, we focus on the rural highlands of Peru, where extreme cold events are most common. The highlands have been left behind by Peru's considerable and sustained economic growth: GDP per capita doubled nationally over the last two decades, yet by 2018 the poverty rate in the rural highlands was 49%, compared with 13.1% in Metropolitan Lima (the Peruvian capital) (INEI, 2019). Compounding this vulnerable situation, the rural highlands are subject to considerable weather shocks — droughts, floods, frosts, and cold waves (World Bank, 2008) — and Peru is one of the countries most vulnerable to climate change (Stern, 2007; Tambet and Stopnitzky, 2021). Frosts and extreme cold events have become increasingly common over the last two decades, affecting millions of Peruvians (Keller and Echeverría, 2013; FAO, 2008), particularly in the highlands, which lie at elevated altitudes (between 500 and 6,798 meters above sea level). In recent years, temperatures have dipped as low as -20°C in some areas, affecting close to 200 thousand inhabitants (Centre for Research on the Epidemiology of Disasters, 2023).

Cold is particularly damaging in this setting. Frosts destroy crops — how much depends on how far and for how long the temperature falls and on the crop's stage of growth (Snyder and Melo-Abreu, 2005) — and the agronomic literature identifies the crops most commonly grown in the highlands, maize and potatoes, as particularly sensitive (Lee and Herbek, 2012; Carter and Hesterman, 1990; Hijmans et al., 2001; Burrows, 2019; Janssen, 2004; Romero et al., 1989).[5] Losses can be extreme: a frost in 2008 destroyed 45% of potato production in several high-altitude provinces (FAO, 2008). Frosts also kill livestock, which households in the Andes hold both as productive assets and as buffer stocks to be sold for cash in times of need (Escobal et al., 1999; Kristjanson et al., 2007; León-Velarde and Quiroz, 2003; Herrero et al., 2013); in 2015, temperatures of -20°C in the department of Puno killed 170 thousand alpacas (BBC News, 2015). The threat is widespread: CENEPRED (2021) estimates that 823 districts, with around 1 million farmers and 3.3 million hectares of agricultural land, are at high or very high risk of frost. Extreme cold also harms health, raising both morbidity (Zhao et al., 2021; Kephart et al., 2022; Sheridan and Allen, 2015) and

[5] Frost-sensitive crops account for the majority of the value of agricultural output in our setting; maize and potatoes alone account for nearly half of households' agricultural revenue (our calculations from the 2014–2018 National Agricultural Survey, Instituto Nacional de Estadística e Informática 2018a).

mortality (Deschenes and Moretti, 2009; Deschênes and Greenstone, 2011), especially among the very young and the elderly and often through respiratory infections (Centers for Disease Control and Prevention, 2024); in 2010, cold related to a *La Niña* event killed 250 children in Peru, mostly under the age of five, from "cold-related respiratory diseases, mostly pneumonia" (Kirkland, 2012; Barbier, 2010).

Severe cold is not a peculiarity of the Peruvian Andes. Despite rising average temperatures, cold waves continue to endanger vulnerable populations, especially in the highland areas of low- and middle-income countries. Cold-related deaths account for nearly ten times more global mortality than heat-related deaths, with the largest burdens in Asia and Africa (Zhao et al., 2021). Recent episodes — the 2021 cold wave in South America, the 2023 cold wave in eastern China, and recurrent agricultural losses in the Himalayas — show that extreme cold remains a serious hazard (Marengo et al., 2023; Ma and Zhu, 2025; Shrestha et al., 2023; Basannagari and Kala, 2013), and its consequences are magnified where housing is poor and resources are scarce.[6] Climate change may even be amplifying such extremes: Arctic warming can destabilize the polar vortex and push frigid air into the mid-latitudes of the Northern Hemisphere (Cohen et al., 2018, 2021; Hamouda et al., 2024), while the La Niña episodes associated with cold events in the Southern Hemisphere are expected to become more frequent and prolonged (Cai et al., 2015; Geng et al., 2023). Cold events have worsened over time in highland regions such as Mongolia (United Nations Development Programme (UNDP), 2025) and Peru's own Mantaro basin (Trasmonte et al., 2008). The consequences of cold that we document below — losses of crops and livestock and illness among the very young and the old — are therefore relevant well beyond Peru.

All in all, the Peruvian highlands are exposed to frosts that damage crops, kill livestock, and harm health, in a country whose political institutions command little trust and whose public services reach the poor unevenly. We ask how such shocks translate into political attitudes and behavior in this setting.

[6] Nor are agricultural losses from frost confined to poor countries: a 2021 frost caused an estimated US$ 1 billion in losses to coffee farmers in the Brazilian state of Minas Gerais (Samora, 2021), and another frost during the same year generated around € 2 billion in losses in French wine production (The Guardian, 2021).

# 3 Data

Our analysis combines five sources: the Peruvian National Household Survey (*Encuesta Nacional de Hogares*, ENAHO), hourly temperature data from the European Centre for Medium-Range Weather Forecasts (ECMWF), district-level results from three presidential elections, the results and candidate lists of four municipal elections, and the AmericasBarometer survey of the Latin American Public Opinion Project (LAPOP). This section describes each source, the outcomes we study, and our estimation sample.

## 3.1 Household survey and outcomes at the individual level

The ENAHO is a cross-sectional household survey collected annually by the National Statistics Office (Instituto Nacional de Estadística e Informática 2018b, INEI). We use twelve rounds, from 2007 to 2018. Fieldwork is continuous throughout each survey year, so that households in the same district are interviewed in different months; we return to this feature of the design in Section 4. The INEI also records where each household lives: in rural areas, the ENAHO reports the GPS coordinates of the village, and in urban areas, the coordinates of the centroid of the neighborhood block.

**Perceptions of democracy.** One randomly chosen adult (18 years or older) in each household answers a module on governance, democracy, and transparency. Our primary outcome comes from the question "In Peru, does democracy work...?", with the response options "very well", "well", "poorly", and "very poorly". We define an indicator that takes the value of one for respondents who believe democracy works well or very well:[7]

$$Y_{idt} = \begin{cases} 1 & \text{if response is "very well" or "well"} \\ 0 & \text{if response is "poorly" or "very poorly"} \end{cases} \tag{1}$$

This question is closely related to the widely used "satisfaction with democracy" (SWD) item,

[7]The questionnaire also includes a "do not know" option, which is chosen by about 24% of respondents. In our main analysis, we drop these observations; Appendix Table A13 shows that our results are robust to treating the full response scale as an ordered outcome and to including "do not know" as a neutral middle category.

which asks respondents how satisfied they are with the way democracy works in their country and which has been interpreted as a combined measure of support for democratic principles and of the evaluation of specific institutions and leaders (Kölln and Aarts, 2021; Clarke et al., 1993; Christmann, 2018; Kosec and Mo, 2023). Rather than choosing between between these interpretations, we treat the belief that democracy works well as an overall evaluation of how democracy delivers, and in Section 6, we use the other items of the module to characterize the effect of cold across different dimensions of political support. These items cover, first, democratic *principles*: whether the respondent considers democracy important for the country and whether democracy is always preferable to any other form of government. Second, democracy *in practice*: whether the respondent believes that, in Peru, there is respect for freedom of expression, equality before the law, political freedom, and transparent elections, each answered on the scale "not at all", "little", "sufficiently", "a lot". We code each item as one for "sufficiently" or "a lot" and summarize the four items as the share of items that the respondent rates as respected, for respondents who answer at least three of the four.[8] Third, confidence in specific *institutions*, on the same four-point scale, coded as one for "sufficiently" or "a lot": the national Congress, political parties, the regional government, the provincial government, and the district/municipality government, as well as three non-political institutions that we use as placebos: the written press, radio and television, and the Catholic church.[9]

**Identity and participation in local associations.** The same module asks the respondent which group they identify with most; we use the answers "my department, province, district or town" and "my indigenous group". From the module on citizen participation we obtain whether any household member participates in a local association, which we group into political and government associations (political parties, municipal management committees, citizen round tables), professional and agricultural associations (worker associations, trade guilds, professional and agricultural associations), and community-based associations (peasant communities in charge

[8] Appendix Table A10 reports the four items separately, as well as two standardized versions of the index (the mean of the standardized items and an inverse-covariance-weighted index in the spirit of Anderson 2008).

[9] The module also asks, from the 2010 round onward, whether the respondent approves of the management of the central, regional, provincial and district governments; Appendix Table A11 reports these items.

of communal land, rural self-defense groups, neighborhood associations).[10]

**Household problems and economic outcomes.** The governance module also asks whether the household faced any of a list of problems during the previous twelve months — a member lost their job, the family business went bankrupt, a member was gravely ill or had an accident, the household suffered a natural disaster — and whether the problem caused the household to lose income or assets. From the agricultural module we obtain each household's agricultural revenue and the value of livestock that died over the previous twelve months, both of which we express in 2007 soles. From the health module, we obtain, for household members aged 5 and under and 65 and over, whether the member suffered an illness in the four weeks before the interview that required a medical consultation, and whether the member was hospitalized. Finally, we use the ENAHO to compile demographic information about the respondent and the household: sex, age, education, mother tongue, household size, and the altitude of the household's location.

### 3.2 Weather data and the cold-shock measure

We collect hourly temperatures from the ERA5 reanalysis of the European Centre for Medium-Range Weather Forecasts (2018) (ECMWF) for every hour (i.e., 1h, 2h, . . . , 23h, 24h) from 2005 through 2018. The ECMWF estimates temperatures from weather stations, satellites, and sondes, and processes this information at a geographic resolution of 0.25 degrees (≈28 km). We match household data from the ENAHO with the ECMWF weather data using households' approximate locations and the month and year of interview. This allows us to construct a household-specific measure of extreme cold exposure throughout the year prior to the interview (the standard recall period for most survey questions).

As described in Bollman et al. (2026), we build on the widely used cumulative degree days measure from Schlenker and Roberts (2006) and estimate the number of cumulative degree *hours* during which a household experienced extreme cold. This measure captures both the duration and

[10] Our measure of participation in local associations aims to capture how citizens self-organize; we therefore exclude participation in government-run social assistance programs (such as *Vaso de Leche* and *Comedores Populares*). Prior to the 2012 round of the ENAHO, participation in *Comunidades Campesinas* (a historical and widespread community-based group) was not asked about directly, but was included in the category "Other". As a result, we include the "Other" category prior to 2012 in the community-based definition.

the severity of extreme cold events — i.e., for how long and by how much a household experienced temperatures below a certain threshold. Denote the temperature threshold $\lambda$, where $\lambda$= 0°C, -1°C, -2°C,... , -12°C. We begin by defining harmful degree hours (DH) as the extent (in °C) to which a given hour's temperature falls below the threshold $\lambda$:

$$\text{Degree Hours}(\text{DH}_{itmdh}) = \begin{cases} \lambda - h_{itmdh} & \text{if } h_{itmdh} < \lambda \\ 0 & h_{itmdh} \geq \lambda \end{cases} \tag{2}$$

where $h_{itmdh}$ is the temperature in household $i$'s location, on year $t$, month $m$, and hour $h$ of day $d$. For example, if $\lambda$=-1°C, an hour of temperature at -3°C represents 2 degree hours. Based on the agronomic literature, we choose a baseline threshold of -9°C, a temperature that is harmful for many crops grown in the highlands (Lee and Herbek, 2012; Carter and Hesterman, 1990; Hijmans et al., 2001; Burrows, 2019; Janssen, 2004; Romero et al., 1989). However, as sensitivity to cold can vary across crops, we show our results using a wide range of temperature thresholds (from 0°C to -12°C).

Our primary measure of extreme cold exposure is the *cumulative* degree hours (CDH) below threshold $\lambda$ that household $i$ interviewed in month $m$ and year $t$ experienced over the 12 months prior to the survey. In other words, we aggregate DH over the hours in the year prior to the interview month.[11]

$$\text{Cumulative Degree Hours}(\text{CDH}_{it}) = \sum_{m=-12}^{-1} \sum_{d=1}^{30} \sum_{h=1}^{24} \text{DH}_{itmdh} \tag{3}$$

Additionally, we extract rainfall data from the Climate Hazards Group InfraRed Precipitation with Station Data, CHIRPS (Funk, 2015). CHIRPS is a global dataset that provides high-resolution rainfall estimates at 0.05 × 0.05-degree resolution. We match rainfall to households using GPS coordinates and interview dates from the ENAHO, using the same procedure as for the temperature data. For the district-level outcomes described below, we measure cold, temperature, and rainfall

[11]We choose this time frame for our baseline CDH calculations to match the reference period of the income and household-problem questions. Appendix Table A12 shows that our results are robust to using shorter windows. For the health outcomes, which have a recall period of four weeks, we measure cold over the eight weeks before the interview.

at the centroid of each district.

### 3.3 Estimation sample

Due to the nature and geographic scope of cold weather shocks in Peru, we restrict our sample to farming households in the highlands.[12] With these restrictions, we obtain a main estimation sample of 57,159 respondents across 938 districts in the Peruvian highlands.[13] Table 1 presents summary statistics for the outcomes, weather variables, and respondent and household characteristics.

About half of the respondents believe that democracy works well in Peru, and only a quarter of the four democracy-in-practice items are rated as respected on average. Confidence in political institutions is low: 10% of respondents express confidence in Congress and 6% in political parties, with somewhat higher confidence in local governments. Adverse events are common: in a typical year, 7.5% of households report that a member was gravely ill or had an accident, 29% report a natural disaster, and 38% report that a problem caused them to lose income or assets. Respondents are predominantly poor and rural: 63% have completed at most primary school, more than half speak Quechua (the most prominent indigenous language in Peru) at home, and the average household lives at an altitude of about 3,200 meters.

In any given year, cold below -9°C is rare: Only 5% of respondents experienced any temperature below -9°C in the 12 months prior to their interview (Appendix Figure A4 shows the share exposed at each threshold), so that the average respondent accumulated less than one harmful degree hour at this threshold, though conditional on being exposed, the average is about 15 degree hours. However, it is not rare for the districts that experience it. Among the districts in our sample exposed at least once, the median district experiences cold below -9°C in five out of the twelve years in our sample period. In a year that a district experiences cold below -9°C, the average CDH is around 22.

[12] Appendix Figure A3a shows that farming households in the Highlands (surveyed in the ENAHO between 2007 and 2018) are located precisely in the regions of Peru that are most susceptible to frosts (Appendix Figure A3b).

[13] There are 1,873 districts in Peru, so our sample covers around half of all districts in the country.

### 3.4 Other data sources

**Organismo Nacional de Procesos Electorales (ONPE)**. We obtain data on voting outcomes at the district level from the Peruvian National Elections Commission (2016). We use the 2006, 2011, and 2016 presidential elections. Presidential elections in Peru proceed in two rounds: in the first round, several candidates compete, and the two runners-up advance to the second round. Our data capture voting participation in both rounds. For each district, we observe the number of registered voters and the number of votes cast, and within the votes cast, how many are valid, blank, or invalid. We match each district and round to the weather at the district centroid during the 12 months before the month of the vote.

**Municipal elections**. To test whether households punish the incumbents they can vote against, we use the district-level results of the 2002, 2006, 2010, and 2014 municipal elections and the lists of candidates and of elected authorities published by the National Elections Board (*Jurado Nacional de Elecciones*, JNE) through its Infogob observatory (Jurado Nacional de Elecciones, 2026). For each district and election, we observe registered voters, votes cast, blank and null ballots, the votes of each list, and the name, organization, and position of every candidate and of the elected mayor. The records carry no identification number, so we identify the incumbent mayor by matching the full name (both surnames and given names) of the mayor elected four years earlier to the mayoral candidates of the current election in the same district. We use the elections up to 2014 because a 2015 law bars mayors from running for immediate re-election, so there are no incumbents on the ballot in the 2018 election. We leave out the 57 district-elections that were annulled on legal grounds (defined by the JNE as having two-thirds or more of the votes cast made up by blank or invalid votes, or fewer than half of registered voters voting; Article 36 of Law 26864).[14] As for the presidential elections, we measure cold at the district centroid over the twelve months before election day.

**Juntos coverage at baseline**. To measure coverage of the state's main cash transfer program in each

[14]We cannot identify the incumbent mayor in 55 of the remaining district-elections. In 50 of these elections, the previous election was annulled on legal grounds and then repeated, making them very different from other regular elections; the other 5 with missing incumbents are the first elections in newly created districts. We exclude elections with missing incumbents from our incumbency regressions but include them in the participation regressions in Appendix Table A16.

province, we use the number of households paid by Juntos in each district and year from 2005 to 2017, published by the program (Ministerio de Desarrollo e Inclusión Social (MIDIS) and Programa Nacional de Apoyo Directo a los más Pobres JUNTOS, nd). We sum the households paid in each province and divide by the number of poor residents of the province, which we compute from the 2007 population census and the poverty rates from the 2007 poverty map of the National Institute of Statistics (Instituto Nacional de Estadística e Informática, 2007, 2009). Dividing by the number of poor residents, rather than by the total population, makes the measure reflect the share of the poor population reached by the program rather than the level of poverty itself. We define high-coverage provinces as those above the median of this measure among the 139 highland provinces. Our main measure uses coverage in 2012; Section 8 explains this choice, and we also report results with coverage in 2007.

**Latin American Public Opinion Project (LAPOP)**. We complement the ENAHO with the AmericasBarometer of the Latin American Public Opinion Project (LAPOP Lab, 2017), a nationally representative survey that asks about satisfaction with democracy, interpersonal trust, voting, and identity. We use the 2006, 2008, 2010, 2012, 2014, and 2017 rounds and, as in our main sample, restrict attention to respondents in the highland strata of the survey. The LAPOP does not release the location of respondents, so we assign each respondent by their municipality (district) and match them to the weather at the district centroid during the 12 months before the month of the interview.

## 4 Empirical Strategy

To estimate the causal effects of extreme cold on political attitudes and behavior, we exploit the variation in exposure generated by the timing of interviews within districts. Specifically, we estimate the following regression:

$$Y_{idmt} = \beta_1 \text{CDH}_{idmt} + \beta_2 Z_{idmt} + \alpha_d + \gamma_t + \theta_m + \varepsilon_{idmt} \tag{4}$$

where $Y_{idmt}$ is an outcome for a randomly chosen adult from household $i$ in district $d$ interviewed in calendar month $m$ of year $t$. $\text{CDH}_{idmt}$ is the number of degree hours below threshold $\lambda$ that the household experienced in the 12-month period before being interviewed. $Z_{idmt}$ is a vector of

predetermined individual and household characteristics (respondent sex, age and age squared, education level, mother tongue, household size, and the altitude of the household's location) as well as weather controls (average temperature and rainfall over the same 12 months).[15] We include fixed effects at the district level $(\alpha_d)$ to account for time-invariant spatial heterogeneity in the incidence of cold shocks and in political attitudes. We also include fixed effects for the interview year $(\gamma_t)$ and calendar month $(\theta_m)$, which account for seasonality and general trends in political attitudes and in cold shocks. For household-level outcomes, $Z$ contains the characteristics of the household head instead of those of the respondent.

The coefficient of interest is $\beta_1$. The underlying identifying assumption is that the incidence and intensity of cold shocks are exogenous with respect to political attitudes, conditional on spatial and temporal fixed effects (and other controls, namely altitude). Though it is possible for households to sort into districts endogenously (for example, poorer households might be concentrated in colder and less desirable areas), we exploit *within-district* variation in the intensity of cold shocks over time. This means that our identifying variation comes from households within the same district who are interviewed at different times — and thus who are subject to different temperature fluctuations that vary exogenously by the date of interview (after netting out general trends and seasonality in weather).[16] As long as households are unable to anticipate fluctuations in the intensity of cold shocks, do not migrate in response to shocks, and the timing of their interviews does not respond to the weather, $\hat{\beta}_1$ captures the causal effect of extreme cold; we examine these conditions in Section 4.1.

Most of the outcomes we examine are binary in nature, so we use a linear probability model and report coefficients and standard errors multiplied by 100. We cluster standard errors at the district level. For our main result, we also report randomization inference $p$-values in Appendix Table A2: under the sharp null hypothesis that cold affects no one, we reassign the district weather series (permuting the survey-year labels of each district's series, or permuting the districts within each survey year) and re-estimate equation (4) one thousand times per scheme.

[15] Table 3 shows that our results are not driven by the inclusion of controls.

[16] The ENAHO is collected continuously throughout each survey year, in a way that allows for areas to be sampled more than once during any given year and provides variation in the survey dates even within a given district.

### 4.1 Threats to identification

**Interview timing.** Our design relies on household interview timing being unrelated to the cold the household has experienced. A concern specific to this design is that fieldwork could respond to the weather conditions, for example, if enumerators avoided the coldest weeks in the coldest districts, thereby mechanically relating the interview month to the shock. Appendix Table A3 tests this directly. We build a panel of districts by survey year and calendar month, and regress the share of the district-year's interviews conducted in each month on the cold recorded at the district centroid in that month, with district, year, and calendar-month fixed effects. Neither the amount of cold nor the incidence of any cold below -9°C in a month changes the share of a district's interviews conducted in it; both coefficients are small and statistically indistinguishable from zero.

**Anticipation and pre-existing trends.** To ensure that our measure of frost shocks captures exogenous weather shocks rather than systematic unobserved determinants or pre-existing trends in perceptions of democracy, we perform a simple falsification test where we estimate the "effect" of future cold weather events. Specifically, we estimate a version of equation (4) where instead of focusing on CDH in the 12 months prior to the survey, we include CDH in the 12 months *after* the interview date.[17] Column 2 of Appendix Table A4 shows no statistically significant relationship between perceptions of democracy and future realizations of extreme cold temperatures. This helps us rule out the possibility that households can anticipate future frost shocks, as well as the potential for frost shocks to capture unobserved determinants of perceptions of democracy that systematically vary across households and geographic areas. It shows that there are no differential pre-trends in perceptions of democracy that are related to frost shocks.

**Migration and sample composition.** Another potential concern is that households may migrate in response to past frost shocks. Such endogenous migration would mean that households who remain in areas experiencing relatively more frost shocks may systematically differ from those living in areas with fewer shocks. Appendix Table A5 shows that the sex, age, household size, and mother tongue of respondents are unrelated to CDH. The exception is education: a one standard

[17] Because we estimate the "effects" of a 12-month lead of CDH and have weather data only through 2018, for this analysis we have a restricted sample period from 2007 to 2017. Thus, we begin by illustrating that our main results hold on this restricted sample period in column 1 of Appendix Table A4.

deviation increase in CDH is associated with a 0.4 percentage point higher probability that the respondent has primary education or less (column 4), on a mean of 63 percent. This difference does not explain our results; respondents with primary education or less are *more* likely, not less, to believe that democracy works well, so a less-educated sample would work against the effect we find. Additionally, in column 2 of Appendix Table A6, we show that the results are robust to restricting the sample to those who are currently living in their district of birth (so-called "non-movers"). We also directly assess whether households move in response to frost shocks and find no evidence that migration status is related to cold shocks (column 3).

## 5 Cold as a Household Shock

Before turning to political attitudes, we establish what cold does to a highland household. Table 2 reports the effects of cold on a set of outcomes that, taken together, describe cold as a multifaceted adverse shock to the household: agricultural revenue and livestock (Panel A), the health of the most vulnerable household members (Panel B), and the problems that households report having faced (Panel C). Each row is a separate regression with the specification of equation (4), estimated on the sample described in the notes to the table. Throughout the paper, we describe magnitudes as the effect of a one standard deviation increase in CDH, which is 11.7 degree hours in our main estimation sample.[18]

**Economic losses.** We find that extreme cold substantially reduces agricultural revenue. The first row of Panel A in Table 2 shows that a one standard deviation increase in CDH in the past year lowers annual agricultural revenue by 1.6%.[19] Extreme cold also harms households' livestock: the same increase translates into a 2.9 percent increase in the value of livestock deaths (second row).[20]

**Health.** We focus on the health of the elderly and the very young, motivated by the recognition that these groups are particularly vulnerable to extreme cold exposure (Kirkland, 2012; Centers for Disease Control and Prevention, 2024; Deschenes and Moretti, 2009). We use two indicators that

[18]When we describe magnitudes, we use the standard deviation in the estimation sample specific to each regression.

[19]We transform monetary outcomes using the inverse hyperbolic sine transformation to interpret the effects of extreme cold in terms of percent changes while accounting for zero-valued observations.

[20]On the extensive margin, the same increase raises the likelihood that the household loses any livestock by 0.5 percentage points (column 1 of Appendix Table A7).

capture the incidence and severity of illness among household members aged 65 and older and children aged 5 and younger: whether the individual suffered from an illness requiring a medical consultation, and whether the individual was hospitalized.[21] Panel B shows that a one standard deviation increase in CDH over the 8 weeks before the interview raises the likelihood of an illness requiring medical attention by 0.4 percentage points and the likelihood of hospitalization by 0.2 percentage points. These are meaningful effects: about 30% of these individuals required medical attention, and 3.4% were hospitalized in the past month, so the effect on hospitalization amounts to a 5-percent increase relative to its mean.[22]

**Problems reported by the household.** Panel C uses the household's own account of the previous twelve months (asked in a separate module from the health and economic questions assessed above). A one standard deviation increase in CDH raises the probability that a household member was gravely ill or had an accident by 0.27 percentage points, a 4 percent increase relative to the mean of 7.5%; the probability that the household suffered a natural disaster by 0.57 percentage points (on a mean of 29%); and the probability that a problem caused the household to lose income or assets by 0.56 percentage points (on a mean of 38%) — all significant at the 1% level. The two items that concern wage employment and non-farm businesses — a member lost their job, the family business went bankrupt — do not respond to cold. We read these null effects with caution: both events are very rare in this population of farmers (1.1% and 0.8% of households report them in a typical year), so there is little room for an effect to appear, and the absence of a response may reflect this floor rather than the absence of a labor-market channel.

Taken together, Table 2 shows that extreme cold is a compound shock: it lowers farm revenue, kills livestock, makes the most vulnerable household members sick, and is experienced by the household as a disaster that costs it income and assets. The following sections ask how households respond to such a shock politically.

[21] Illness is reported for the 4 weeks prior to the survey interview. We match the household's weather data from the 8 weeks prior to the interview date to allow for extreme cold conditions to affect health with a lag.

[22] We pool the two age groups for precision. Appendix Table A8 shows the results separately for children aged 5 and under and adults aged 65 and over: the effect on illness requiring medical attention is present in both groups and is largest for young children, while the effect on hospitalization is concentrated among the elderly.

# 6 Effects of Cold on Perceptions of Democracy

Section 5 established that extreme cold reaches the household through its farm, its animals, and the health of its most vulnerable members. We now turn to the central question of the paper: how do households that experience such a shock judge the political system? We begin with our primary outcome, the belief that democracy works well, and then characterize effects across various dimensions of political support.

We find that cold weather shocks worsen individuals' evaluation of democracy. A one standard deviation increase in CDH in the year before the interview reduces the probability that an individual believes democracy works well (or very well) by 0.40 percentage points (Table 3, column 1). Results using our preferred specification, which also includes the individual and household controls, are very similar: 0.44 percentage points (column 2), or 0.9 percent of the sample mean. Exposure for the average household in any given year is rare, so this average effect is small. However, the effect is sizable for those who live through a severe winter; exposure of 100 degree hours — about three nights at -12°C (10°F), the level experienced by households in our sample during severe winters, and roughly the cumulative exposure for the average exposed district over our sample period — lowers the belief that democracy works well by nearly 4 percentage points (7 percent relative to the mean).

**Magnitudes.** Two kinds of benchmark put these numbers in perspective. The first is the variation in perceptions across groups within our sample, which our regressions hold fixed: women are 3.9 percentage points more likely than men to believe that democracy works well, and respondents who completed secondary school are about 7 percentage points less likely to believe democracy works well than those who do not.[23] The effect of a severe winter (as defined above) is about the size of the gender gap and about half the gap between respondents with secondary and primary schooling. The second benchmark is the response of satisfaction with democracy and political trust to other adverse conditions. Blanco and Ruiz (2013) estimate that having been the victim of a crime in the previous year lowers satisfaction with democracy in Colombia by roughly 5

[23] These gaps are the coefficients on the respondent controls in the specification of column 2 of Table 3, with district, year, and month fixed effects.

percentage points in the probability of being satisfied for a respondent at the median; a severe winter thus has about three-quarters of the effect of crime victimization. Stevenson and Wolfers (2011) estimate that a one percentage point rise in a country's unemployment rate lowers the share of respondents who express confidence in the national government by 1.6 percentage points, so that a severe winter lowers the evaluation of democracy by about as much as a two-point rise in unemployment lowers confidence in government. Ahlerup et al. (2024) find that a one standard deviation increase in drought conditions in Africa lowers trust in the president by about 3 percent of a standard deviation, and a severe drought by about 9 percent; the corresponding figures here are 0.9 percent of a standard deviation for one standard deviation of cold and 8 percent for a severe winter.

**Robustness.** The result is not sensitive to how the shock or the outcome is measured, or to how the fixed effects are specified. The effect grows monotonically as we consider more extreme temperature thresholds (Appendix Figure A6); it is somewhat larger when CDH is aggregated over the 6 or 3 months before the interview, and it is present when we use cumulative degree days, an indicator for any cold below the threshold (which lowers positive perceptions by 3.2 percentage points), a winsorized, log, or inverse hyperbolic sine transformation of CDH, or CDH measured at the district centroid rather than at the household's location (Appendix Table A12). Retaining the full response scale in an ordered probit, or including the "do not know" responses as a neutral middle category, gives the same picture (Appendix Table A13). Replacing the separate year and month effects with year-by-month effects, which absorb any shock common to all households interviewed in the same month of the same year, gives a similar estimate; adding district-by-year and district-by-month effects, so that the effect is identified only from households in the same district and survey year interviewed in different months, net of each district's own seasonal pattern, gives a slightly larger and more precise one; and replacing district fixed effects with conglomerate (primary sampling unit) fixed effects, which are finer than Peru's already small districts, gives a larger estimate still (Appendix Table A14).[24] Finally, the effect is not specific to the ENAHO: in the LAPOP, whose respondents are matched to district-level weather, exposure to cold in the year

[24]Our sample includes 2,985 conglomerates and 938 districts.

before the interview also lowers satisfaction with democracy (Appendix Table A1, column 1).

**Which level of support?** The belief that democracy works well could fall because respondents turn against democracy as a principle, because they lose confidence in particular bodies, or because they judge that the regime is delivering less. The other items of the governance module let us tell these apart. Figure 1 plots the effect of cold on each item, estimated with the specification of column 2 of Table 3; the coefficients are reported in Appendix Table A9. Panel (a) shows that the shock touches evaluations of how democracy works but not democratic principles. Alongside the belief that democracy works well, the democracy-in-practice index falls: a one standard deviation increase in CDH reduces the share of the four rights that respondents rate as respected by 0.34 percentage points, on a mean of 25%, and each of the four components moves in the same direction (Appendix Table A10). By contrast, the two items that capture support for democracy as a principle — that democracy is important for the country and that democracy is always preferable to any other form of government — do not show statistically significant movements; their point estimates are positive, and their confidence intervals exclude declines of the size found for the belief that democracy works well.

Panel (b) shows that the shock does not single out any government institution for blame. Confidence in Congress, political parties, and the regional, provincial, and district government is unaffected, with point estimates close to zero. This could in part reflect floors: only 10% of respondents express confidence in Congress and 6% in political parties, so there is little room to fall. However, we also observe no effects on respondents' confidence in regional, provincial and district governments, which have higher baseline means and thus are less likely to be subject to floor effects. Similarly, the shock has no significant effect on the approval of the way the central, regional, provincial governments are managed (Appendix Table A11), despite also having much higher baseline means, though there is a slight increase in approval of district government (column 4, significant at the 10% level). The effect of extreme cold therefore sits at the level of regime performance, between principles and institutions: households hit by cold do not abandon democracy as an ideal, and they do not blame a particular government or body; they conclude that democracy is delivering less — that it works less well and that the rights it promises are less respected. This pattern foreshadows the behavioral responses in Section 7: a judgment about the system as a whole, rather than about an actor, is more likely to appear as withdrawal than as a

change in whom one supports.

If a frost simply soured respondents' moods or changed how they answered any evaluative survey question, confidence in institutions unrelated to the state should fall as well. Appendix Figure A5 shows that it does not: the effects of cold on confidence in the written press, in radio and television, and in the Catholic church are all close to zero and precisely estimated. The decline is specific to the evaluation of how democracy delivers.

# 7 Effects of Cold on Electoral Participation, Local Associations, and Identity

Extreme cold worsens the evaluation of how democracy works. What are the behavioral consequences? We next examine three outcomes: participation in elections, participation in local associations, and the group with which respondents identify. Ex ante, it is not clear whether extreme cold will increase or decrease voter participation. On the one hand, as cold events erode support for the current government, citizens may express discontent by voting against the incumbent party, leaving total voter turnout unchanged or even increasing it. On the other hand, if extreme cold worsens perceptions of the broader democratic system, disillusioned citizens may refrain from engaging in elections altogether, and turnout may fall.

One feature of the Peruvian setting matters for how we read the electoral results: punishment of an incumbent cannot be observed in presidential elections. The constitution bars immediate presidential re-election, and in each of the three elections we study, the governing party's candidate withdrew before the vote, so that neither a sitting president nor the governing party was on the ballot. Our presidential outcomes therefore capture participation rather than the reallocation of votes among candidates. Municipal elections, in which incumbent mayors do stand for re-election, provide the direct test of punishment, and we turn to them after the presidential results.

Moreover, individuals may seek additional support from informal, community-based institutions as an adaptive coping strategy to mitigate the negative effects of extreme cold. Prior work suggests that in the face of economic shocks, households use local, community-based strategies to cope with economic uncertainties and consumption losses when high-level state and market-led opportunities for consumption smoothing are lacking (Bhattamishra and Barrett,

2010; Buggle and Durante, 2021). Thus, cold exposure may increase engagement in local, informal, community-based associations, and, to the extent that these associations are organized around ethnic and communal identities, strengthen those identities.

**Participation in presidential elections.** To estimate the effects of extreme cold on electoral participation, we use district-level data from both rounds of voting in the 2006, 2011, and 2016 presidential elections. We consider two outcomes separately for each round: the share of absent voters (i.e., the share of registered voters in a district who did not cast a vote) and the combined share of absent voters and blank votes (i.e., votes that are cast but left blank), which have been used as a measure of "protest" votes (Alvarez et al., 2018; Cohen, 2017). We measure CDH at the district centroid over the 12 months before each round of voting,[25] include district and election-year fixed effects, and weight districts by their number of registered voters.[26]

Extreme cold in the year leading up to a presidential election reduces voter participation. A one standard deviation increase in CDH raises the share of absent voters by 0.28 percentage points in the first round and by 0.30 percentage points in the second round (Table 4, columns 1 and 2), increases of about 1.7 percent relative to the mean absentee shares of 16% and 19%. The results are similar when blank votes are counted as non-participation (columns 3 and 4).[27] Though the effect size is modest, it is not negligible relative to the margins by which national elections are decided in Peru: the second round of the 2016 presidential election was decided by 41,057 votes, about 0.2 percent of registered voters, less than the increase in the absentee share that a one standard deviation increase in cold produces in an exposed district. These results align with recent findings in other contexts, where awareness of weak public service delivery reduces the likelihood of voting (Cox et al., 2024). They are also consistent with the reading of Section 6: a judgment about how the system as a whole is delivering, rather than about a particular actor, shows up as withdrawal from elections rather than as a reallocation of votes among candidates.[28]

[25] By construction, CDH does not include weather in the month of the election, so that we do not capture the effects of any cold events that occur on election day that may directly affect voters' ability or willingness to go to the polls.

[26] The results are similar without weights (Appendix Table A15).

[27] The LAPOP points the same way at the individual level: respondents exposed to cold in the year before a presidential election are more likely to report that they did *not* vote in it (Appendix Table A1, column 3).

[28] Here, our results differ from (Balcazar and Kennard, 2025), who find that heat shocks raise turnout in state elections and lower the incumbent party's probability of re-election. However, the context differs in important ways that explain this disparate findings: India is a setting where state governments run agricultural policy and the incumbent party is on

**Municipal elections: incumbent mayors and participation.** Municipal elections let us test punishment directly. Using the JNE's records of candidates and elected authorities (Section 3.4), we determine, for each district in the 2006, 2010, and 2014 elections, whether the mayor elected four years earlier ran again, whether they were re-elected, and the share of the vote their list obtained, and we relate these outcomes to CDH at the district centroid over the twelve months before the vote with the specification of Table 4. Incumbent mayors are not punished (Appendix Table A16). Cold does not lower the probability that the incumbent runs again, which is 56 percent on average, nor the probability that they are re-elected or the share of the vote they receive; if anything, the point estimates are positive, and none is statistically distinguishable from zero.[29] Participation generally responds as in presidential elections: a one standard deviation increase in CDH raises the share of absent voters by about 0.15 percentage points in the weighted specification, though the effect on the share of absent and blank votes is smaller and not statistically significant. Thus overall, results from the municipal elections confirm the pattern in presidential elections: households withdraw from the polls, and they do not use the vote to punish the incumbents they could remove.

**Participation in local associations.** To investigate whether extreme cold shocks affect engagement in local institutions, we evaluate households' participation in the three categories of associations described in Section 3.1: political and government, professional and agricultural, and community-based. Overall participation in local associations rises by 0.28 percentage points for a one standard deviation increase in CDH (Table 5, column 1). This effect is not driven by participation in political and government associations (column 2) or in professional and agricultural associations (column 3), but by community-based associations — peasant communities, rural self-defense groups, and neighborhood associations — where participation rises by 0.36 percentage points (column 4).[30]

**Identity.** The turn toward community-based associations is mirrored in how respondents describe themselves. The governance module asks respondents which group they identify with most. A

the ballot. In Peru's presidential elections, neither condition holds.

[29] Incumbency is defined for the person rather than the party label, because two-thirds of the mayors who run again do so under a different organization.

[30] In the LAPOP, exposure to cold in the year before the interview is also associated with a higher probability of reporting that people in the community are trustworthy, although the estimate is imprecise (Appendix Table A1, column 2). Buggle and Durante (2021) document a similar relationship between historical climate variability and interpersonal trust in Europe.

one standard deviation increase in CDH lowers the probability that the respondent names their department, province, district, or town by 0.35 percentage points (Table 5, column 5; mean 44%) and raises the probability that they name their indigenous group by 0.56 percentage points (column 6; mean 38%). The LAPOP shows the same shift: respondents exposed to cold in the year before the interview are more likely to self-identify as indigenous (Appendix Table A1, column 4). In the highlands, the indigenous identity is the one around which the main community-based institutions — above all the *comunidades campesinas* — are organized, so the two results describe the same movement: away from the administrative units of the state and toward the community.

Overall, these results indicate that extreme cold shocks reduce participation in formal, national institutions (such as voting) and increase engagement in, and identification with, local and informal institutions related to kinship, community, and social proximity.

## 8 Heterogeneous Effects by Baseline Coverage of Cash Transfers

Does the political response to a cold shock depend on how far the state's safety net reaches? Section 5 showed that cold reaches households through farm income, livestock, and the health of their most vulnerable members. Where a cash transfer reaches more of the poor, households may be better able to absorb such losses and the state is visibly present when the shock arrives, so a shock may be less likely to be read as evidence that the system is failing them.[31] We explore this possibility by allowing the effects of cold to differ with the baseline coverage of Juntos, Peru's flagship conditional cash transfer program.

We measure coverage as the number of households paid by Juntos per 100 *poor* residents of the province (Section 3) and compare provinces above and below the median. Our main measure uses coverage in 2012. Juntos expanded in the highlands in two main waves (Section 2): by 2007 it paid households in 46 percent of highland districts and reached a further fifth of them in 2011–2012 (covering 66 percent of districts in 2012) and expanded more slowly thereafter. Coverage in 2012 therefore describes the program's reach over most of the period in which we measure outcomes,

[31] Several studies find that redistributive programs such as cash transfers can directly foster support for government (Evans et al., 2019; Kosec and Mo, 2023), though effects may be short-lived (Zucco Jr, 2013).

while coverage in 2007 reflects only the first wave and classifies as low-coverage several provinces that were covered in subsequent years. For the household outcomes, we use the 2013–2018 rounds of the ENAHO, after coverage is measured. For the electoral outcomes, we use the 2011 and 2016 presidential elections; we exclude the 2006 election because its outcome could have influenced where the program expanded in the following year. Juntos selects poor districts, so we measure coverage relative to the number of poor residents rather than to the whole population, in order to keep the split from simply ranking provinces by poverty.

Table 6 reports the results. Column 1 shows that the effect of cold on the belief that democracy works well in the 2013–2018 sample is nearly identical to the full-sample estimate of Table 3. Column 2 allows the effect to differ by coverage. A one standard deviation increase in CDH lowers the probability that a respondent believes democracy works well by 1.2 percentage points in provinces with low coverage and has no statistically significant effect in provinces with high coverage; the difference in effects across high and low coverage provinces is significant at the 95 percent level of confidence. Cold raises absenteeism in both groups of provinces, but by more in those with low Juntos coverage, though only the second-round difference is statistically significant. The shift toward indigenous identification is smaller in provinces with high coverage, in line with the results for democracy and voting, but the difference is not statistically significant (column 6), and the increase in participation in community-based associations does not differ significantly by coverage (column 5). We thus find no evidence that community-based institutions substitute for the program (cf. Banerjee et al., 2024).

Appendix Table A17 reports two alternative specifications. Panel B measures coverage in 2007, before the second wave of the program, and uses the 2008–2018 rounds of the ENAHO. The results are very similar: cold lowers the belief that democracy works well in provinces with low coverage but not in those with high coverage (the interaction is 0.080, significant at the 5 percent level), and the second-round increase in absenteeism is again smaller where coverage is high. Panel C returns to coverage in 2012 and adds the interaction of CDH with the province poverty rate. The interaction between CDH and the indicator for above-median Juntos coverage is positive for democracy evaluation and negative for voting, community associations, and indigenous identity (aligning with expectations), though it is only statistically significant for democracy evaluation.

Taken together, the decline in the belief that democracy works well is concentrated in areas where

the state's main cash transfer program reaches few of the poor. This is consistent with transfers acting as insurance: where the program reaches more households, a cold shock is less likely to be read as evidence that democracy fails to deliver. Similarly, the effect of cold on absenteeism is weaker in areas with high cash transfer coverage. However, because coverage is not randomly assigned, we regard these patterns as suggestive.

## 9 Conclusion

When a shock reveals that the state does not protect them, citizens can reject the political system, punish those in office, or withdraw. This paper uses extreme cold in the Peruvian highlands — an exogenous shock that lowers farm revenue, kills livestock, and causes household members to fall ill — to ask which of these responses households choose. We find a specific pattern. Households conclude that democracy is delivering less: they become less likely to believe that democracy works well and that the rights it promises are respected. They do not abandon democracy as a principle; they do not blame any particular government or institution; and, where they can punish an incumbent in municipal elections, they do not. And they withdraw: absenteeism rises in presidential and municipal elections, participation in community-based associations rises, and respondents identify less with the state's administrative units and more with their indigenous group.

Two features of this pattern deserve emphasis. First, the decline in support sits at the level of regime performance, between democratic principles and specific institutions. This is the level at which the widely used "satisfaction with democracy" question operates, and our results show that it responds to material conditions even when principles and institutional confidence do not. Second, the behavioral response is "exit" rather than "voice". Where incumbents can be punished in municipal elections, they are not; the electoral response, in national and local elections alike, is withdrawal from the polls, which is costly in a country where abstention is fined, and it comes together with a turn toward the local institutions and identities that help households cope with the next shock.

Where Juntos, the state's main cash transfer program, reached more of the poor, the decline in the belief that democracy works well disappears and the increase in absenteeism is smaller.

Coverage is not randomly assigned across provinces and we treat this evidence as suggestive, but it points in the same direction as work showing that redistributive programs sustain support for government: the political cost of a shock is lower where the state is visibly present when the shock arrives. For governments in vulnerable regions, this suggests that social protection that reaches households when extreme weather hits protects not only livelihoods but also the relationship between citizens and the democratic system.

Our estimates capture responses within a year of the shock. Whether the withdrawal we document persists, accumulates over repeated shocks, or reverses when the state responds are questions for future work, as is whether the strengthening of local institutions and identities that we observe is a source of resilience or of fragmentation. What our results establish is that extreme cold, a hazard that climate change may make more common in highland regions of the developing world, has consequences beyond livelihoods and health: it changes how citizens judge the system that governs them and how much they take part in it.

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Table 1: Descriptive Statistics

| | Mean (1) | Std. Dev. (2) | Observations (3) |
|---|---|---|---|
| *Perceptions of democracy (respondents to the governance module)* | | | |
| Believes democracy works well | 0.511 | 0.500 | 57,159 |
| Democracy-in-practice index (share of 4 items respected) | 0.247 | 0.316 | 57,079 |
| Democracy is important | 0.840 | 0.367 | 61,877 |
| Democracy preferable to autocracy | 0.706 | 0.456 | 57,925 |
| *Confidence in political institutions* | | | |
| Congress | 0.103 | 0.303 | 55,414 |
| Political parties | 0.061 | 0.240 | 60,515 |
| Regional government | 0.138 | 0.345 | 57,045 |
| Provincial municipality | 0.173 | 0.379 | 63,940 |
| District municipality | 0.207 | 0.405 | 67,845 |
| *Problems the household reports for the last 12 months* | | | |
| A household member lost their job | 0.011 | 0.106 | 75,847 |
| The family business went bankrupt | 0.008 | 0.087 | 75,847 |
| A household member was gravely ill or had an accident | 0.075 | 0.263 | 75,847 |
| The household suffered a natural disaster | 0.288 | 0.453 | 75,847 |
| The household lost income or assets because of a problem | 0.382 | 0.486 | 75,846 |
| *Agricultural income, livestock, and health (household and member level)* | | | |
| Agricultural revenue (2007 soles) | 2746.5 | 7616.3 | 76,642 |
| Value of livestock deaths (2007 soles) | 131.1 | 484.1 | 63,025 |
| Required medical attention (members aged 5 and under or 65 and over) | 0.302 | 0.459 | 63,084 |
| Was hospitalized (members aged 5 and under or 65 and over) | 0.034 | 0.182 | 63,076 |
| *Participation in local associations (household level) and identity* | | | |
| Participates in any local association | 0.816 | 0.387 | 76,471 |
| Participates in a political or government association | 0.012 | 0.107 | 76,471 |
| Participates in a professional or agricultural association | 0.182 | 0.386 | 76,471 |
| Participates in a community-based association | 0.703 | 0.457 | 76,471 |
| Identifies most with department, province, district or town | 0.439 | 0.496 | 75,625 |
| Identifies most with indigenous group | 0.376 | 0.484 | 75,625 |
| *Electoral participation (district level, 2006, 2011 and 2016 elections)* | | | |
| Share of absent voters, first round | 0.197 | 0.091 | 3,813 |
| Share of absent voters, second round | 0.223 | 0.099 | 3,813 |
| Share of absent and blank votes, first round | 0.344 | 0.094 | 3,813 |
| Share of absent and blank votes, second round | 0.235 | 0.098 | 3,813 |
| *Weather (main estimation sample)* | | | |
| CDH below -9°C, 12 months before the interview (degree hours) | 0.755 | 11.682 | 57,159 |
| Any cold below -9°C in the 12 months before the interview | 0.050 | 0.219 | 57,159 |
| Average temperature (°C) | 9.542 | 3.379 | 57,159 |
| Average monthly rainfall (mm) | 65.303 | 21.610 | 57,159 |
| *Respondent and household characteristics (main estimation sample)* | | | |
| Male | 0.509 | 0.500 | 57,159 |
| Age | 46.724 | 16.911 | 57,159 |
| Household size | 4.014 | 2.056 | 57,159 |
| Education: primary or less | 0.628 | 0.483 | 57,159 |
| Education: secondary | 0.285 | 0.451 | 57,159 |
| Education: technical | 0.050 | 0.217 | 57,159 |
| Education: college | 0.037 | 0.190 | 57,159 |
| Mother tongue: Quechua | 0.518 | 0.500 | 57,159 |
| Mother tongue: Spanish | 0.428 | 0.495 | 57,159 |
| Mother tongue: Aymara or other indigenous | 0.055 | 0.227 | 57,159 |
| Altitude (meters) | 3185.7 | 692.8 | 57,159 |

Notes: Individual-level outcomes, weather and characteristics refer to respondents to the ENAHO governance module living in farming households in the Highlands, 2007-2018; each outcome is summarized on the sample of its regression. The main estimation sample is the sample of Table 3. Agricultural revenue and the value of livestock deaths are for households with agricultural revenue over the previous year, in 2007 soles (deflated with the GDP deflator). Health outcomes refer to household members aged 5 and under or 65 and over. Participation in local associations is measured at the household level. Electoral outcomes are unweighted district-level shares of registered voters. CDH is the sum over the 12 months before the interview of the number of degrees below -9°C in each hour; it is divided by 100 in the regressions.

Table 2: Effects of Frost Shocks on Household Income, Assets, Health, and Reported Problems

| | Coefficient on CDH ($\lambda$ = -9°C) (1) | Mean of Dep. Var. (2) | Observations (3) | No. of Districts (4) |
|---|---|---|---|---|
| *Panel A: Agricultural income and livestock (households with agricultural activity)* | | | | |
| Agricultural revenue | -0.135*** | 2746.5 | 76,642 | 944 |
| | (0.042) | | | |
| Value of livestock deaths | 0.252* | 131.1 | 63,025 | 922 |
| | (0.141) | | | |
| *Panel B: Health of household members aged 5 and under or 65 and over* | | | | |
| Required medical attention | 0.059*** | 0.302 | 63,084 | 937 |
| | (0.014) | | | |
| Was hospitalized | 0.029*** | 0.034 | 63,076 | 937 |
| | (0.004) | | | |
| *Panel C: Problems the household reports for the last 12 months* | | | | |
| A household member lost their job | -0.001 | 0.011 | 75,847 | 942 |
| | (0.002) | | | |
| The family business went bankrupt | -0.002 | 0.008 | 75,847 | 942 |
| | (0.003) | | | |
| A household member was gravely ill or had an accident | 0.023*** | 0.075 | 75,847 | 942 |
| | (0.008) | | | |
| The household suffered a natural disaster | 0.049*** | 0.288 | 75,847 | 942 |
| | (0.018) | | | |
| The household lost income or assets because of a problem | 0.048*** | 0.382 | 75,846 | 942 |
| | (0.014) | | | |

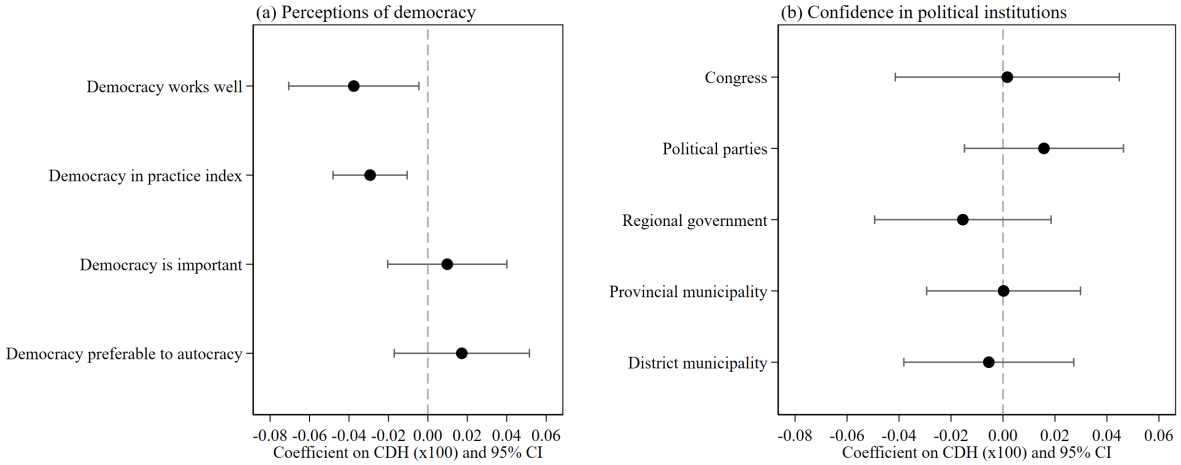
Notes: Each row is a separate regression of the outcome on cumulative degree hours below -9°C over the 12 months before the interview (CDH). Panel A: households in the Highlands with agricultural revenue over the previous year, ENAHO 2007-2018; outcomes are transformed with the inverse hyperbolic sine, and the mean of the dependent variable is in 2007 soles. Controls include average temperature and rainfall at the household level over the same period, altitude, household head characteristics (sex, age, age squared, education level and mother tongue fixed effects), log of total land (owned and rented), and household size fixed effects. Panel B: individuals aged 5 and under or 65 and over living in farming households in the Highlands, ENAHO 2007-2018; controls include average temperature and rainfall, altitude, the individual's sex and age fixed effects, household head characteristics and household size fixed effects. Panel C: respondents to the governance module in farming households in the Highlands, ENAHO 2007-2018; the outcomes are indicators for problems the household reports having faced in the last 12 months, and the last row equals one if a problem caused the household to lose income or assets. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 3: Effects of Frost Shocks on the Belief that Democracy Works Well

| | Dep. Var.: Believes Democracy Works Well | |
|---|---|---|
| | Only Weather Controls | All Controls |
| | (1) | (2) |
| Cumulative Degree Hours ($\lambda$ = -9°C) | -0.034** | -0.038** |
| | (0.017) | (0.017) |
| Observations | 57159 | 57159 |
| No. of Districts | 938 | 938 |
| Mean of Dep. Var | 0.511 | 0.511 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. All specifications include year, district, and month of interview fixed effects as well as average temperature and average rainfall at the household level over the same reference period as the frost shock. Column 2 additionally includes controls for individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), household size fixed effects, and altitude. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Figure 1: Effects of Frost Shocks Across Levels of Political Support




Notes: Each marker is the coefficient on cumulative degree hours below -9°C over the 12 months before the interview (CDH) from a separate regression, with its 95 percent confidence interval based on district-clustered standard errors. The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. Panel (a): believes democracy works well or very well; the democracy-in-practice index is the share of four items (freedom of expression, equality before the law, political freedom, transparent elections) that the respondent rates as sufficiently or fully respected; democracy is important; democracy is always preferable to autocracy. Panel (b): confidence (sufficient or a lot) in each institution. Coefficients and standard errors have been multiplied by 100, as in the tables; the coefficients are reported in Appendix Table A9.

Table 4: Effects of Frost Shocks on Electoral Participation

| | Share of Absent Votes | | Share of Absent & Blank Votes | |
|---|---|---|---|---|
| | First Round | Second Round | First Round | Second Round |
| | (1) | (2) | (3) | (4) |
| CDH ($\lambda$ = -9°C) | 0.010*** | 0.010*** | 0.008*** | 0.011*** |
| | (0.002) | (0.002) | (0.001) | (0.002) |
| Observations | 3813 | 3813 | 3813 | 3813 |
| No. of Districts | 1278 | 1278 | 1278 | 1278 |
| Mean of Dep. Var | 0.164 | 0.187 | 0.285 | 0.197 |

Notes: Shares are calculated with respect to the total eligible voters in each district. Weather variables are measured at the district centroid and measure weather in the year prior to the date of each round of voting. The sample includes all districts in the Highlands and covers the 2006, 2011 and 2016 presidential elections. All specifications include year and district fixed effects. Observations are weighted by the number of registered voters in each district and election round. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 5: Effects of Frost Shocks on Participation in Local Associations and on Identity

| | Participation in local associations | | | | Identifies most with ... | |
|---|---|---|---|---|---|---|
| | All Local Assoc. | Political & Government | Professional & Agricultural | Community-Based | Dept., Province, District, or Town | Indigenous Group |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| CDH ($\lambda$ = -9°C) | 0.024** | 0.000 | 0.004 | 0.031** | -0.030* | 0.048*** |
| | (0.010) | (0.001) | (0.017) | (0.014) | (0.017) | (0.015) |
| Observations | 76471 | 76471 | 76471 | 76471 | 75625 | 75625 |
| No. of Districts | 944 | 944 | 944 | 944 | 940 | 940 |
| Mean of Dep. Var | 0.816 | 0.012 | 0.182 | 0.703 | 0.439 | 0.376 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Columns 1-4 are at the household level: the outcome equals one if any household member participates in an association of the type indicated (government-run social assistance programs are excluded); controls include average temperature, average rainfall and altitude at the household level over the same reference period as the frost shock, household head characteristics (sex, age, age squared, education level, and mother tongue), and household size fixed effects. Columns 5-6 are at the individual level: the outcome equals one if the respondent to the governance module says the group they identify most with is their department, province, district or town (column 5) or their indigenous group (column 6); controls include average temperature, average rainfall and altitude, respondent sex, age and age squared, education level and mother tongue fixed effects, and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

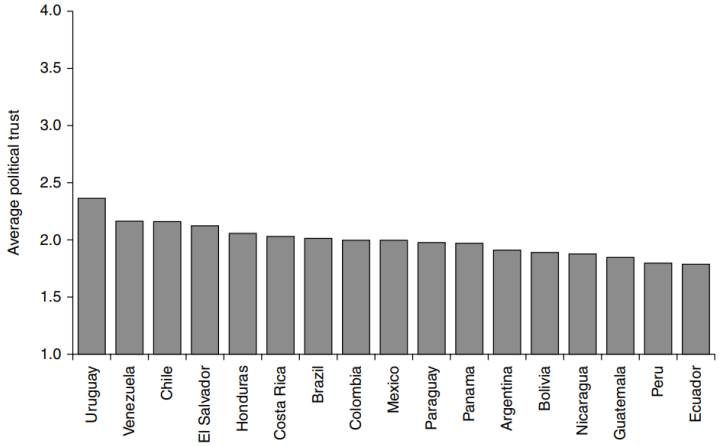
Table 6: Heterogeneous Effects by Juntos Coverage

| | Democracy Works Well | | Share of Absent Voters | | Participates in Community-Based Associations | Identifies Most with Indigenous Group |
|---|---|---|---|---|---|---|
| | | | First Round | Second Round | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| CDH ($\lambda$ = -9°C) | -0.036** | -0.103*** | 0.010*** | 0.017*** | -0.019 | 0.075** |
| | (0.018) | (0.040) | (0.003) | (0.004) | (0.051) | (0.032) |
| CDH × High Juntos Coverage | | 0.089** | -0.002 | -0.008** | 0.044 | -0.029 |
| | | (0.042) | (0.003) | (0.004) | (0.051) | (0.035) |
| Total Effect for High Juntos Coverage Provinces | | -0.014 | 0.008*** | 0.009*** | 0.025*** | 0.046*** |
| | | (0.014) | (0.001) | (0.001) | (0.007) | (0.014) |
| Observations | 34,295 | 34,295 | 2,546 | 2,546 | 43,875 | 43,323 |
| No. of Districts | 890 | 890 | 1,278 | 1,278 | 897 | 892 |
| Mean of Dep. Var | 0.555 | 0.555 | 0.187 | 0.211 | 0.702 | 0.378 |

Notes: Juntos coverage is measured in 2012; ENAHO outcomes (columns 1-2, 5 and 6) are from 2013-2018. Juntos coverage is the number of households paid by the Juntos conditional cash transfer program (administrative records of the program, summed over the districts of the province) per 100 poor residents of the province (2007 census population times the 2007 INEI district poverty rate, summed over the districts of the province). High Juntos Coverage equals one for provinces above the median across the 139 highland provinces. Columns 1-2: individuals in all farming households in the Highlands; controls include average temperature, average rainfall and altitude at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, age squared, education level, and mother tongue), and household size fixed effects; year, district, and month of interview fixed effects. Columns 3-4: shares of absent voters with respect to eligible voters in each district; all districts in the Highlands, 2011 and 2016 presidential elections; weather at the district centroid in the year prior to each round; year and district fixed effects; observations weighted by registered voters. Column 5: household participation in a community-based association, with the household-level controls of Table 5. Column 6: the respondent identifies most with their indigenous group, with the controls of columns 1-2. Total effect is the sum of the CDH coefficient and the interaction. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

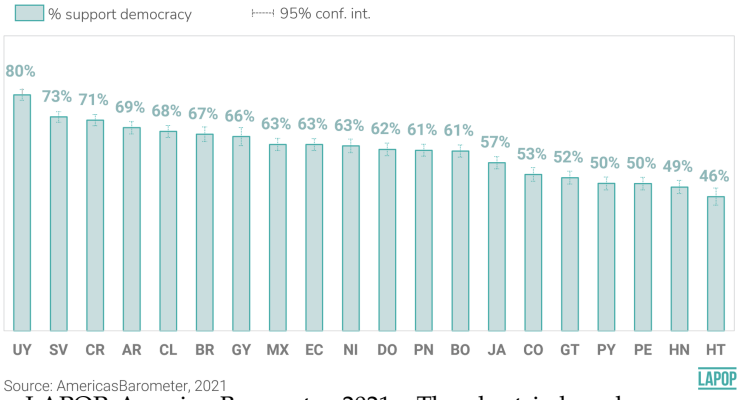
# A Appendix: Figures and Tables

Figure A1: Trust in Political Institutions in Latin America, 1996-2011



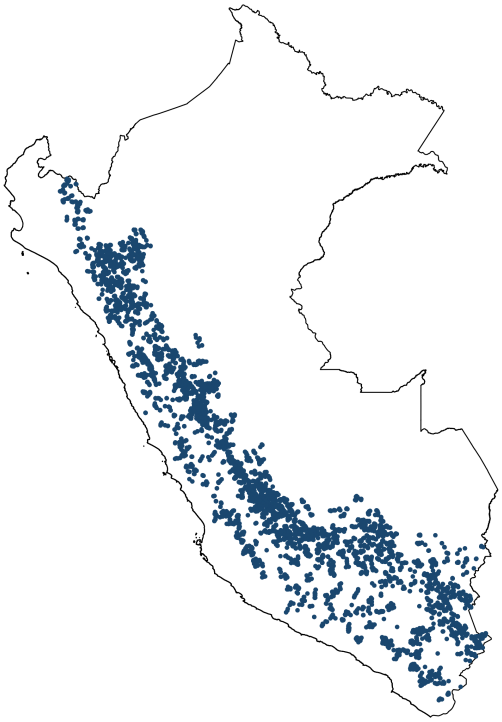

Source: Bargsted, Somma and Castillo (2017), Handbook of Political Trust. The index is an average of trust in three political institutions (the national congress, political parties, and the government), averaged over the period 1996-2011.

Figure A2: Preferences for Democracy in Latin America, 2021

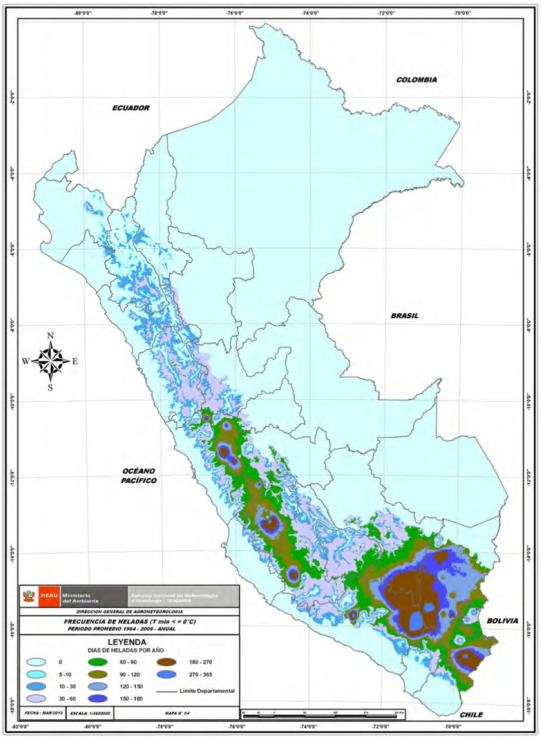



Source: LAPOP AmericasBarometer 2021. The chart is based on respondents' answers to the question *Democracy may have problems, but it is better than other forms of government. To what extent do you agree with this statement?* Responses ranged from one (strongly disagree) to seven (strongly agree); those answering five, six, or seven are considered to support democracy.

Figure A3: Location of the ENAHO Households in This Study and the Incidence of Frosts in Peru



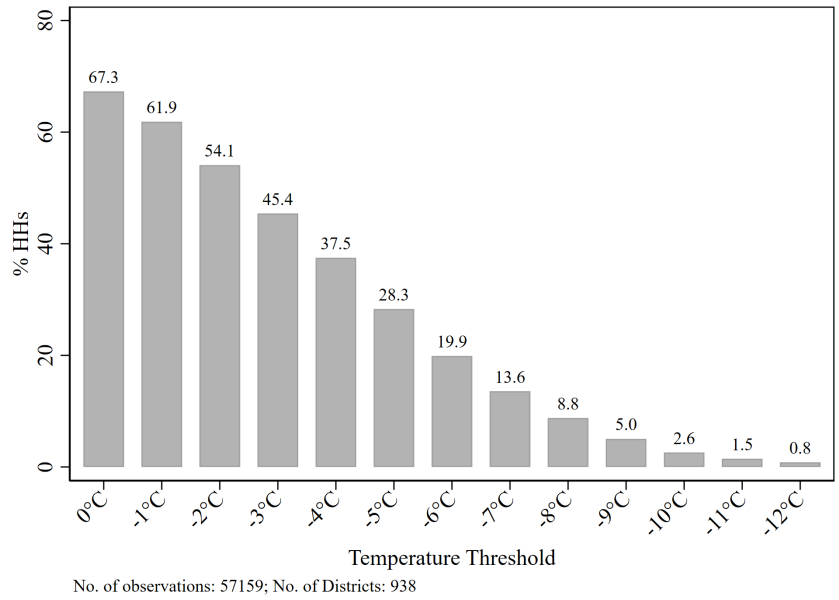
(a) Location of the households in this study, ENAHO 2007-2018




(b) Incidence of frosts in Peru, 1964-2009

Notes: The left panel presents the location of the sample of farming households in the Peruvian Highlands used in this study (ENAHO 2007-2018). The right panel is taken from SENAMHI (2010, p. 47) and presents the average number of frost days per year between 1964 and 2009.

Figure A4: Percentage of Households Facing Cold Below Each Temperature Threshold




Notes: Share of the main estimation sample (individuals in farming households in the Highlands, ENAHO 2007-2018) whose household experienced at least one degree hour below the threshold indicated during the 12 months before the interview.

Table A1: Frost Shocks and Political Attitudes in the LAPOP, Highlands 2006–2017

| | Satisfied with democracy (1) | Trusts people in community (2) | Did not vote in last presidential election (3) | Self-identifies as indigenous (4) |
|---|---|---|---|---|
| CDH ($\lambda$ = -9°C) | -0.158** | 0.176 | 2.286*** | 0.140*** |
| | (0.076) | (0.120) | (0.541) | (0.050) |
| Observations | 2430 | 2782 | 1491 | 2652 |
| No. of districts | 85 | 85 | 84 | 85 |
| Respondents with any frost | 138 | 161 | 102 | 152 |
| Mean of dep. var | 0.342 | 0.455 | 0.064 | 0.234 |
| Rounds | 2006-2017 | 2006-2017 | 2006-2017 | 2006-2017 |

Notes: Respondents in the three Sierra strata of the AmericasBarometer (LAPOP) Peru, rounds 2006, 2008, 2010, 2012, 2014 and 2017. Weather is measured at the centroid of the respondent's district over the 12 months before the interview month in columns 1, 2 and 4, and over the 12 months before the first round of the 2006, 2011 and 2016 presidential elections in column 3 (rounds 2006, 2012 and 2017; respondents aged 18 at the interview are left out, since most of them were too young to vote in that election). All columns include district, survey-round and interview-month fixed effects (district and election-year fixed effects in column 3), average temperature and rainfall over the same window, sex, age, age squared and education-level fixed effects. Column 4 equals one for indígena in 2006–2012 and quechua, aymara or amazónico in 2014–2017. CDH is divided by 100. District-clustered standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A2: Randomization Inference for the Effect of Frost Shocks on the Belief that Democracy Works Well

| | Dep. Var.: Believes Democracy Works Well | |
|---|---|---|
| | CDH at the household location | CDH at the district centroid |
| | (1) | (2) |
| CDH ($\lambda$ = -9°C) | -0.038** | -0.027*** |
| | (0.017) | (0.009) |
| p-value, district-clustered standard errors | 0.026 | 0.004 |
| *Randomization inference p-values (1000 permutations each)* | | |
| Survey years permuted within district | – | 0.082 |
| Districts permuted within survey year | – | 0.083 |
| Observations | 57,159 | 57,159 |
| No. of Districts | 938 | 938 |
| Mean of Dep. Var | 0.511 | 0.511 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Column 1 is the main specification of Table 3, column 2. Column 2 measures cold, average temperature and rainfall at the centroid of the household's district over the same 12 months before the interview month, so that whole district weather series can be reassigned. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. Randomization inference: under the null hypothesis that cold affects no one, the weather series are reassigned and the specification of column 2 is re-estimated 1000 times per scheme; the p-value is the share of permuted coefficients at least as large in absolute value as the observed one. The first scheme permutes the survey-year labels of each district's weather series independently across districts, keeping each household's interview month; the second reassigns, within each survey year, the weather series of the sample districts at random across districts. The permuted coefficients are centred near zero (means -0.001 and -0.005, standard deviations 0.016 and 0.015). District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A3: Interview Timing: Does Cold in a Month Change the Share of a District's Interviews Conducted in It?

| | Dep. Var.: Share of the district-year's interviews conducted in the month | |
|---|---|---|
| | (1) | (2) |
| District CDH in the month ($\lambda$ = -9°C) | 0.008 | |
| | (0.018) | |
| Any cold below -9°C in the month | | -0.007 |
| | | (0.009) |
| District-months | 91488 | 91488 |
| District-years | 7624 | 7624 |
| No. of Districts | 938 | 938 |
| Mean of Dep. Var | 0.083 | 0.083 |

Notes: One observation per district, survey year and calendar month, for every district-year in which at least one household of the main estimation sample was interviewed (months without interviews enter as zeros). The dependent variable is the number of main-sample households interviewed in the district in the month divided by the number interviewed in the district in the year. Cold is cumulative degree hours below -9°C at the district centroid in the calendar month, divided by 100 (column 1), or an indicator for any such cold (column 2). Both columns include district, year, and calendar-month fixed effects. District-level clustered standard errors in parentheses. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A4: Falsification Test: Effects of Future Frost Shocks

| | Dependent Variable: Believes Democracy Works Well | |
|---|---|---|
| | (1) | (2) |
| CDH ($\lambda$ = -9°C) in the *Previous* 12 Months | -0.043*** | |
| | (0.016) | |
| CDH ($\lambda$ = -9°C) in the *Next* 12 Months | | 0.011 |
| | | (0.012) |
| Observations | 50701 | 50701 |
| No. of Districts | 902 | 902 |
| Mean of Dep. Var | 0.515 | 0.515 |

Notes: The sample includes all individuals in farming households in the Highlands using the 2007-2017 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), altitude, and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A5: Frost Shocks and Sample Composition

| | Male | Age | Household Size | Primary Education | Speaks Quechua |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| CDH ($\lambda$ = -9°C) | 0.002 | -0.122 | 0.049 | 0.034*** | 0.002 |
| | (0.013) | (0.315) | (0.045) | (0.012) | (0.009) |
| Observations | 57159 | 57159 | 57159 | 57159 | 57159 |
| No. of Districts | 938 | 938 | 938 | 938 | 938 |
| Mean of Dep. Var | 0.509 | 46.724 | 4.014 | 0.628 | 0.518 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Except when used as an outcome, controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, age squared, education level fixed effects, mother tongue); altitude, and household size. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A6: Assessing Endogenous Migration

| | Dep. Var.: Believes Democracy Works Well | | Dep. Var.: Migrated |
|---|---|---|---|
| | Full Sample | Non-movers | Full Sample |
| | (1) | (2) | (3) |
| CDH ($\lambda$ = -9°C) | -0.038** | -0.033* | -0.009 |
| | (0.017) | (0.018) | (0.010) |
| Observations | 57159 | 46907 | 57151 |
| No. of Districts | 938 | 913 | 938 |
| Mean of Dep. Var | 0.511 | 0.518 | 0.179 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO; in column 2, the sample is further restricted to individuals who reside in their district of birth (non-movers). Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, age squared, education level fixed effects, mother tongue), altitude, and household size. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A7: Effects of Frost Shocks on Livestock Assets

| | Any Livestock Death (1) | Log Value of Livestock Deaths (2) |
|---|---|---|
| CDH ($\lambda$ = -9°C) | 0.045* | 0.252* |
| | (0.025) | (0.141) |
| Observations | 63028 | 63025 |
| No. of Districts | 922 | 922 |
| Mean of Dep. Var | 0.339 | 131.098 |

Notes: Value of livestock deaths has been transformed using the inverse hyperbolic sine function. The sample includes all farming households using the 2007-2018 rounds of the ENAHO. Controls include average temperature, average rainfall, altitude at the household level for over the same reference period as the frost shock, household head characteristics (sex, age, and age squared, education level and mother tongue), log of total land (owned + rented), and household size. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1. Mean value of livestock deaths are expressed in 2007 soles using the GDP deflator published by World Bank (2023).

Table A8: Effects of Frost Shocks on Health

| | Individual Required Medical Attention | | Individual Was Hospitalized | |
|---|---|---|---|---|
| | Age $\leq 5$ (1) | Age $\geq 65$ (2) | Age $\leq 5$ (3) | Age $\geq 65$ (4) |
| Cumulative Degree Hours ($\lambda$ = -9°C) | 0.199*** | 0.044*** | 0.028 | 0.029*** |
| | (0.066) | (0.014) | (0.029) | (0.003) |
| Observations | 29242 | 33811 | 29240 | 33805 |
| No. of Districts | 887 | 918 | 887 | 918 |
| Mean of Dep. Var | 0.311 | 0.295 | 0.022 | 0.045 |

Notes: The sample includes all individuals 5 and under (columns 1 and 3) or individuals 65 and over (columns 2 and 4) living in farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude, household head characteristics (sex, age, and age squared as well as education level and mother tongue fixed effects), individual sex and age fixed effects, and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A9: Effects of Frost Shocks Across Levels of Political Support: Coefficients Behind Figure 1

| | Coefficient on CDH ($\lambda$ = -9°C) (1) | Mean of Dep. Var. (2) | Observations (3) | No. of Districts (4) |
|---|---|---|---|---|
| *Panel A: Perceptions of democracy* | | | | |
| Democracy works well | -0.038** | 0.511 | 57,159 | 938 |
| | (0.017) | | | |
| Democracy in practice index | -0.029*** | 0.247 | 57,079 | 939 |
| | (0.010) | | | |
| Democracy is important | 0.010 | 0.840 | 61,877 | 939 |
| | (0.015) | | | |
| Democracy preferable to autocracy | 0.017 | 0.706 | 57,925 | 938 |
| | (0.017) | | | |
| *Panel B: Confidence in political institutions* | | | | |
| Congress | 0.002 | 0.103 | 55,414 | 939 |
| | (0.022) | | | |
| Political parties | 0.016 | 0.061 | 60,515 | 940 |
| | (0.016) | | | |
| Regional government | -0.015 | 0.138 | 57,045 | 939 |
| | (0.017) | | | |
| Provincial municipality | 0.000 | 0.173 | 63,940 | 939 |
| | (0.015) | | | |
| District municipality | -0.005 | 0.207 | 67,845 | 940 |
| | (0.017) | | | |

Notes: Each row is a separate regression with the main specification. The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. Panel A: believes democracy works well or very well; the democracy-in-practice index is the share of four items (freedom of expression, equality before the law, political freedom, transparent elections) rated as sufficiently or fully respected, for respondents answering at least three of the four; democracy is important; democracy is always preferable to autocracy. Panel B: confidence (sufficient or a lot) in the institution. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A10: Effects of Frost Shocks on the Democracy-in-Practice Index and Its Components

| | Index | | | Components: believes there is respect for ... | | | |
|---|---|---|---|---|---|---|---|
| | Mean of items | Mean of z-scores | Inverse-cov. weighted | Freedom of expression | Equality before the law | Political freedom | Transparent elections |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| CDH ($\lambda$ = -9°C) | -0.029*** | -0.094*** | -0.094*** | -0.029*** | -0.022** | -0.038* | -0.022 |
| | (0.010) | (0.030) | (0.030) | (0.010) | (0.009) | (0.020) | (0.016) |
| Observations | 57079 | 57079 | 57079 | 58245 | 60187 | 58831 | 56263 |
| No. of Districts | 939 | 939 | 939 | 939 | 939 | 937 | 939 |
| Mean of Dep. Var | 0.247 | 0.000 | 0.000 | 0.233 | 0.151 | 0.351 | 0.251 |

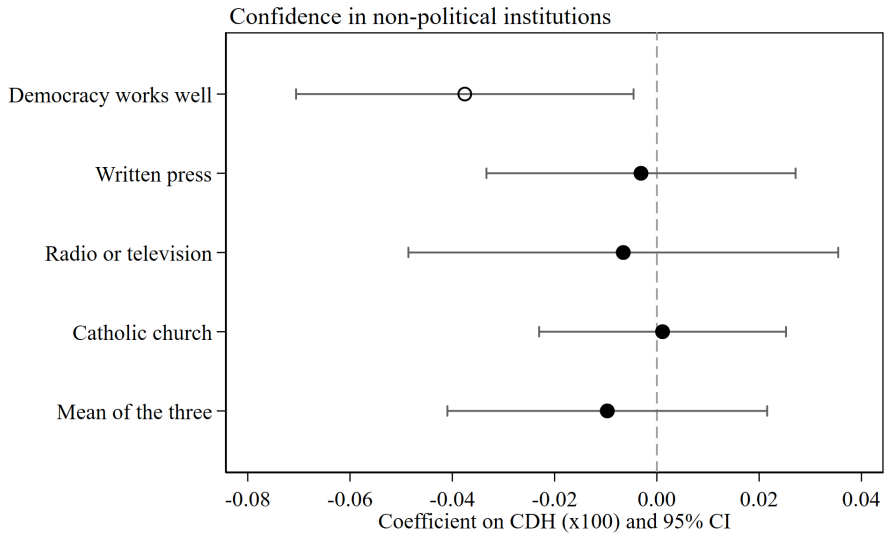
Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. The governance module asks whether, in Peru, there is respect for freedom of expression, equality before the law, political freedom, and transparent elections, with the answers nada, poco, suficiente, bastante; each component in columns 4-7 equals one for suficiente or bastante. Column 1 is the share of the four items rated suficiente or bastante (respondents answering at least three of the four); column 2 is the mean of the standardized items, standardized; column 3 weights the standardized items by the inverse of their covariance matrix (Anderson, 2008), standardized. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A11: Effects of Frost Shocks on Approval of Government Management

| | Approves of the management of the ... government | | | |
|---|---|---|---|---|
| | Central | Regional | Provincial | District |
| | (1) | (2) | (3) | (4) |
| CDH ($\lambda$ = -9°C) | 0.015 | 0.036 | 0.014 | 0.029* |
| | (0.015) | (0.036) | (0.022) | (0.017) |
| Observations | 46335 | 40513 | 44856 | 49976 |
| No. of Districts | 932 | 928 | 933 | 933 |
| Mean of Dep. Var | 0.519 | 0.440 | 0.454 | 0.472 |
| ENAHO rounds | 2010-2018 | 2010-2018 | 2010-2018 | 2010-2018 |

Notes: Individuals in all farming households in the Highlands. The questions on the management of the central and regional governments are asked from the 2010 round of the ENAHO; the provincial and district governments are asked about jointly as the local government in 2010-2011 and separately from 2012 onwards (both are used). The outcome equals one when the respondent rates the management as good or very good and zero when the respondent rates it as bad or very bad. Columns 1 and 2 exclude the interviews conducted between January and March of 2010: in those months the questions on the central and regional governments also offered the answer "regular", which was removed from the questionnaire in April 2010, so that the answers given in those months are not comparable with those given afterwards. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Figure A5: Placebo Outcomes: Confidence in Non-Political Institutions



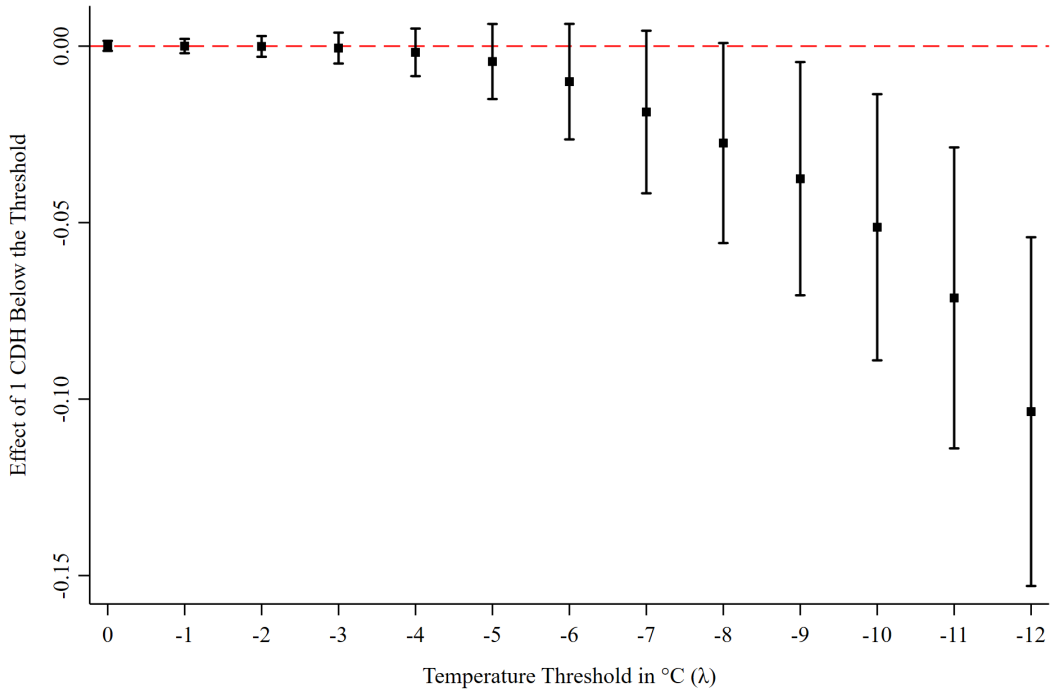

Notes: Each marker is the coefficient on CDH below -9°C over the 12 months before the interview from a separate regression with the main specification (see Figure 1), with its 95 percent confidence interval based on district-clustered standard errors; the hollow marker reproduces the belief that democracy works well for reference. Outcomes equal one when the respondent reports sufficient or a lot of confidence in the institution; the last row is the mean of the three. Coefficients (standard errors) multiplied by 100: Democracy works well: -0.038** (0.017), N = 57,159; Written press: -0.003 (0.015), N = 56,719; Radio or television: -0.007 (0.021), N = 65,569; Catholic church: 0.001 (0.012), N = 71,225; Mean of the three: -0.010 (0.016), N = 66,048;

Table A12: Effects Using Alternative Measures of Frost Shocks

| | Dep. Var.: Believes Democracy Works Well | | | |
|---|---|---|---|---|
| | Coefficient on the shock measure (1) | Mean of Dep. Var. (2) | Observations (3) | No. of Districts (4) |
| CDH over the past 12 months (baseline) | -0.038** | 0.511 | 57,159 | 938 |
| | (0.017) | | | |
| CDH over the past 6 months | -0.055*** | 0.511 | 57,159 | 938 |
| | (0.018) | | | |
| CDH over the past 3 months | -0.040*** | 0.511 | 57,159 | 938 |
| | (0.015) | | | |
| Cumulative degree days over the past 12 months | -0.179* | 0.511 | 57,159 | 938 |
| | (0.103) | | | |
| Any cold below the threshold over the past 12 months (0/1) | -3.206* | 0.511 | 57,159 | 938 |
| | (1.674) | | | |
| CDH winsorized at the 99th percentile | -0.435* | 0.511 | 57,159 | 938 |
| | (0.223) | | | |
| log(1 + CDH) | -1.518** | 0.511 | 57,159 | 938 |
| | (0.761) | | | |
| asinh(CDH) | -1.239** | 0.511 | 57,159 | 938 |
| | (0.631) | | | |
| CDH measured at the district centroid | -0.027*** | 0.511 | 57,159 | 938 |
| | (0.009) | | | |

Notes: Each row is a separate regression of the belief that democracy works well on the shock measure indicated, all at the threshold of -9°C. The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Rows 2 and 3 aggregate degree hours over the 6 and 3 months before the interview and control for average temperature over the same window; row 4 uses cumulative degree days; row 5 an indicator for any degree hour below the threshold; row 6 caps CDH at its 99th percentile; rows 7 and 8 transform CDH with the natural log and the inverse hyperbolic sine; row 9 measures CDH, average temperature and rainfall at the centroid of the household's district. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Figure A6: Effects of Frost Shocks on the Belief that Democracy Works Well, by Temperature Threshold




Observations: 57159; No. of Districts: 938
Mean of Dep. Var=0.51

Notes: Each marker is the coefficient on cumulative degree hours below the threshold indicated on the horizontal axis, over the 12 months before the interview, from a separate regression with the main specification, with its 95 percent confidence interval. The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, altitude at the household level, individual characteristics (respondent sex, age, and age squared as well as education level and mother tongue fixed effects), and household size fixed effects. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors. Coefficients and standard errors have been multiplied by 100 for ease of interpretation.

Table A13: Ordered Probit Models (Marginal Effects)

| | A. Dep. Var.: Believes Democracy Works Well (categories) | | | | |
|---|---|---|---|---|---|
| | Very Poorly | Poorly | Don't Know | Well | Very Well |
| | (1) | (2) | (3) | (4) | (5) |
| CDH ($\lambda$ = -9 °C) | 0.007** | 0.015** | – | -0.019** | -0.003** |
| | (0.003) | (0.007) | – | (0.009) | (0.001) |
| N. of obs. | | | 57167 | | |

| | B. Dep. Var.: Believes Democracy Works Well (categories) | | | | |
|---|---|---|---|---|---|
| | Very Poorly | Poorly | Don't Know | Well | Very Well |
| | (6) | (7) | (8) | (9) | (10) |
| CDH ($\lambda$ = -9 °C) | 0.005* | 0.014* | 0.0002* | -0.017* | -0.002* |
| | (0.003) | (0.008) | (0.0001) | (0.009) | (0.001) |
| N. of obs. | | | 75632 | | |

Notes: Columns 1-5: Dependent variable is categorical and takes 4 distinct values (omitting "Don't Know"). Columns 6-10: Dependent variable is categorical and takes 5 distinct values (including "Don't Know" as a "middle" category). The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Controls include average temperature and average rainfall at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, age squared, education level, and mother tongue), altitude, and fixed effects for household size. All specifications include year, district, and month of interview fixed effects. District-level clustered standard errors in parentheses. Marginal Effects and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A14: Robustness to Alternative Fixed Effects

| | Dep. Var.: Believes Democracy Works Well | | | |
|---|---|---|---|---|
| | Baseline | Year × month FE | District × year and district × month FE | Conglomerate FE |
| | (1) | (2) | (3) | (4) |
| CDH ($\lambda$ = -9°C) | -0.038** | -0.031* | -0.041*** | -0.065*** |
| | (0.017) | (0.017) | (0.012) | (0.019) |
| Observations | 57159 | 57159 | 56671 | 56823 |
| No. of Districts | 938 | 938 | 923 | 932 |
| Mean of Dep. Var | 0.511 | 0.511 | 0.512 | 0.512 |

Notes: The sample includes individuals in all farming households in the Highlands using the 2007-2018 rounds of the ENAHO. Main specification and sample; only the fixed effects change. Column 1: district, year and month-of-interview fixed effects. Column 2: district and year-by-month fixed effects. Column 3: district-by-year fixed effects (which also absorb district-specific trends) and district-by-month fixed effects (district-specific seasonality), so that the effect is identified from households in the same district and survey year interviewed in different months. Column 4: conglomerate (primary sampling unit) fixed effects instead of district fixed effects, with year and month fixed effects. Household-level weather controls and individual controls as in the main specification. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A15: Effects of Frost Shocks on Electoral Participation: Unweighted Estimates

| | Share of Absent Votes | | Share of Absent & Blank Votes | |
|---|---|---|---|---|
| | First Round | Second Round | First Round | Second Round |
| | (1) | (2) | (3) | (4) |
| CDH ($\lambda$ = -9°C) | 0.007*** | 0.009*** | 0.006*** | 0.009*** |
| | (0.002) | (0.003) | (0.002) | (0.003) |
| Observations | 3803 | 3803 | 3803 | 3803 |
| No. of Districts | 1268 | 1268 | 1268 | 1268 |
| Mean of Dep. Var | 0.197 | 0.223 | 0.344 | 0.235 |

Notes: Same specification as Table 4, without weighting districts by their number of registered voters (the sample differs only by ten district-election observations that are singletons once the weights are removed): shares are calculated with respect to the total eligible voters in each district, weather variables are measured at the district centroid in the year prior to the date of each round of voting, the sample includes all districts in the Highlands in the 2006, 2011 and 2016 presidential elections, and all specifications include year and district fixed effects. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A16: Effects of Frost Shocks on Municipal Elections: Incumbent Mayors and Participation

| | CDH ($\lambda$ = -9°C) | | Mean of | | No. of |
|---|---|---|---|---|---|
| | Weighted | Unweighted | Dep. Var. | Observations | Districts |
| | (1) | (2) | (3) | (4) | (5) |
| Incumbent mayor ran again | 0.162 | 0.260* | 0.563 | 3,286 | 1,104 |
| | (0.201) | (0.148) | | | |
| Incumbent mayor re-elected, conditional on running | -0.011 | 0.242 | 0.315 | 1,511 | 651 |
| | (0.239) | (0.202) | | | |
| Incumbent mayor re-elected (0 if did not run) | 0.122 | 0.173 | 0.175 | 3,285 | 1,104 |
| | (0.112) | (0.117) | | | |
| Incumbent mayor's vote share (0 if did not run) | 0.081 | 0.077 | 0.150 | 3,286 | 1,104 |
| | (0.059) | (0.047) | | | |
| Incumbent mayor's vote share, conditional on running | 0.025 | 0.003 | 0.270 | 1,512 | 651 |
| | (0.046) | (0.047) | | | |
| Share of absent voters | 0.019*** | 0.012* | 0.143 | 3,378 | 1,145 |
| | (0.007) | (0.006) | | | |
| Share of absent voters and blank votes | 0.007 | 0.004 | 0.221 | 3,378 | 1,145 |
| | (0.011) | (0.011) | | | |

Notes: Each cell in columns 1-2 is a separate regression of the outcome in the row on CDH below -9°C at the district centroid over the twelve months before the October municipal election (Oct t-1 to Sep t, divided by 100, as in Table 4), average temperature and average monthly rainfall over the same window, with district and election-year fixed effects; all districts in the Highlands, municipal elections of 2006, 2010 and 2014 (a 2015 law bars the immediate re-election of mayors, so the 2018 election has no incumbents on the ballot). The 57 district-elections that meet the legal grounds for annulment (null and blank votes, together or separately, above two thirds of the votes cast, or fewer than half of registered voters casting a ballot; Article 36 of Law 26864) are left out. Column 1 weights districts by registered voters; column 2 is unweighted. Columns 3-5 refer to the unweighted sample. The incumbent mayor is the person elected district mayor four years earlier, from the JNE's records of elected authorities, matched by full name to the mayoral candidates of the current election in the same district (the records carry no identification number). Rows conditional on running use the districts in which the incumbent mayor was a candidate. Absent voters are registered voters who did not cast a ballot; the last row adds blank ballots; both are shares of registered voters. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.

Table A17: Heterogeneous Effects by Juntos Coverage: Alternative Specifications

| | Democracy Works Well | | Share of Absent Voters | | Participates in Community-Based Associations | Identifies Most with Indigenous Group |
|---|---|---|---|---|---|---|
| | | | First Round | Second Round | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A. Main specification (Table 6): 2012 baseline, ENAHO 2013-2018* | | | | | | |
| CDH ($\lambda$ = -9°C) | -0.036** | -0.103*** | 0.010*** | 0.017*** | -0.019 | 0.075** |
| | (0.018) | (0.040) | (0.003) | (0.004) | (0.051) | (0.032) |
| CDH × High Juntos Coverage | | 0.089** | -0.002 | -0.008** | 0.044 | -0.029 |
| | | (0.042) | (0.003) | (0.004) | (0.051) | (0.035) |
| Total Effect for High Juntos Coverage Provinces | | -0.014 | 0.008*** | 0.009*** | 0.025*** | 0.046*** |
| | | (0.014) | (0.001) | (0.001) | (0.007) | (0.014) |
| Observations | 34,295 | 34,295 | 2,546 | 2,546 | 43,875 | 43,323 |
| No. of Districts | 890 | 890 | 1,278 | 1,278 | 897 | 892 |
| Mean of Dep. Var | 0.555 | 0.555 | 0.187 | 0.211 | 0.702 | 0.378 |
| *Panel B. 2007 baseline, ENAHO 2008-2018* | | | | | | |
| CDH ($\lambda$ = -9°C) | -0.037** | -0.097** | 0.010*** | 0.016*** | 0.005 | 0.058* |
| | (0.017) | (0.038) | (0.003) | (0.004) | (0.042) | (0.030) |
| CDH × High Juntos Coverage | | 0.080** | -0.002 | -0.007** | 0.028 | -0.018 |
| | | (0.040) | (0.003) | (0.004) | (0.043) | (0.033) |
| Total Effect for High Juntos Coverage Provinces | | -0.017 | 0.008*** | 0.009*** | 0.033*** | 0.040*** |
| | | (0.012) | (0.001) | (0.001) | (0.007) | (0.013) |
| Observations | 53,696 | 53,696 | 2,546 | 2,546 | 71,289 | 70,475 |
| No. of Districts | 938 | 938 | 1,278 | 1,278 | 944 | 940 |
| Mean of Dep. Var | 0.521 | 0.521 | 0.187 | 0.211 | 0.709 | 0.378 |
| *Panel C. 2012 baseline, controlling for CDH × province poverty rate* | | | | | | |
| CDH ($\lambda$ = -9°C) | -0.036** | -0.325*** | 0.012*** | 0.018*** | 0.092 | 0.127 |
| | (0.018) | (0.064) | (0.004) | (0.004) | (0.068) | (0.167) |
| CDH × High Juntos Coverage | | 0.480*** | -0.009 | -0.011 | -0.144 | -0.116 |
| | | (0.108) | (0.009) | (0.010) | (0.105) | (0.273) |
| Total Effect for High Juntos Coverage Provinces | | 0.155*** | 0.003 | 0.006 | -0.052 | 0.010 |
| | | (0.048) | (0.007) | (0.007) | (0.047) | (0.108) |
| Observations | 34,295 | 34,295 | 2,546 | 2,546 | 43,875 | 43,323 |
| No. of Districts | 890 | 890 | 1,278 | 1,278 | 897 | 892 |
| Mean of Dep. Var | 0.555 | 0.555 | 0.187 | 0.211 | 0.702 | 0.378 |

Notes: Panel A repeats Table 6: Juntos coverage in 2012, ENAHO outcomes from 2013-2018. Panel B measures Juntos coverage in 2007, with ENAHO outcomes from 2008-2018. Panel C repeats Panel A and adds, in columns 2-6, the interaction of CDH with the province poverty rate (2007 INEI poverty map, population-weighted mean of the district rates; coefficient not reported), centered at its mean in each estimation sample. Juntos coverage is the number of households paid by the Juntos conditional cash transfer program (administrative records of the program, summed over the districts of the province) per 100 poor residents of the province (2007 census population times the 2007 INEI district poverty rate, summed over the districts of the province). High Juntos Coverage equals one for provinces above the median across the 139 highland provinces. Columns 1-2: individuals in all farming households in the Highlands; controls include average temperature, average rainfall and altitude at the household level over the same reference period as the frost shock, individual characteristics (respondent sex, age, age squared, education level, and mother tongue), and household size fixed effects; year, district, and month of interview fixed effects. Columns 3-4: shares of absent voters with respect to eligible voters in each district; all districts in the Highlands, 2011 and 2016 presidential elections; weather at the district centroid in the year prior to each round; year and district fixed effects; observations weighted by registered voters. Column 5: household participation in a community-based association, with the household-level controls of Table 5. Column 6: the respondent identifies most with their indigenous group, with the controls of columns 1-2. Total effect is the sum of the CDH coefficient and the interaction. District-level clustered standard errors in parentheses. Coefficients and standard errors have been multiplied by 100 for ease of interpretation. Significance levels denoted by: *** p<0.01, ** p<0.05, * p<0.1.